BEFORE THE
STATE OF NEW YORK
PUBLIC SERVICE COMMISSION

In the Matter of

The Brooklyn Union Gas Company d/b/a National Grid NY &
KeySpan Gas East Corporation d/b/a National Grid

Cases 23-G-0225 & 23-G-0226

September 1, 2023

Prepared Testimony of:

Kwaku Duah
Principal Utility Financial
Analyst

Office of Accounting, Audits,
and Finance
State of New York
Department of Public Service
Three Empire State Plaza
Albany, New York 12223-1350

Q. Please state your name, employer, and business address.
A. My name is Kwaku Duah, and I am employed by the New York State Department of Public Service, referred to as the Department. My business address is Three Empire State Plaza, Albany, New York 12223.
Q. Mr. Duah, what is your position at the Department?
A. I am employed as a Principal Utility Financial Analyst in the Office of Accounting, Audits, and Finance.
Q. Please describe your educational background and professional experience.
A. I received my master's degree in Business Administration with a concentration in Finance and Accounting from the State University of New York Institute of Technology, now known as SUNY Polytechnic Institute, in 2005. Before joining the Department, I interned for Destiny USA in Syracuse as a financial analyst. In that role, I assisted with analyzing and validating the financial model of the company's projects. I joined the Department in January 2007 as a Senior

Utility Financial Analyst.

Q. Please briefly describe your current responsibilities with the Department.

A. As a Principal Utility Financial Analyst, I analyze the financial condition, financing mechanisms, risk, costs of debt and equity, diversification, and the relative competitive position of utilities and their holding company parent(s). My assignments generally involve testifying in rate cases on financial issues, analyzing financing proposals, and working on special projects that usually involve risk and return analyses.

Q. Have you previously testified in a regulatory proceeding before the New York State Public Service Commission, referred to as the Commission?

A. Yes, I have testified before the Commission in the following rate cases: Cases 09-E-0715 and 09-G-0716, New York State Electric & Gas Corporation, or NYSEG; Cases 09-E-0717 and 09-G-0718, Rochester Gas and Electric Corporation, or RG&E; Cases 08-G-0609, 10-E-0500, 12-E-0201 and 12-G-0202, Niagara Mohawk Power Corporation d/b/a

National Grid; Case 11-W-0020, Long Island Water Corporation; Cases 13-W-0539, 13-W-0564 and 14-W-0006, United Water New Rochelle and United Water Westchester; Cases 14-E-0318, 14-G-0319, 17-E-0459 and 17-G-0460, Central Hudson Gas and Electric Corporation, or Central Hudson; Cases 16-W-0130, 19-W-0168, 19-W-0269, Suez Water New York; Case 16-W-0259, New York American Water Company, Inc.; Case 19-G-0309, The Brooklyn Union Gas Company d/b/a National Grid NY, referred to as KEDNY; Case 19-G-0310, KeySpan Gas East Corporation d/b/a National Grid, referred to as KEDLI; and Case 21-G-0577, Liberty Utilities (St. Lawrence Gas) Corp.

Purpose of Testimony

Q. What is the purpose of your testimony in these proceedings?

A. The purpose of my testimony is to recommend the fair rate of return to be used by the Staff Accounting Panel to determine the revenue requirements for KEDNY and KEDLI, collectively referred to as the Companies. My testimony will concentrate on the appropriate ratemaking capital structure, as well as the various cost of capital

components, that I recommend be used for the 12-month period ending March 31, 2025, or the Rate Year. I will also respond to the Companies' Capital Structure Panel and the testimony of Joshua Nowak, the Companies' consultant for return on common equity. Additionally, I will discuss the Companies' proposed rate of return for the National Grid USA Service Company's, referred to as Service Company, asset recovery charge, which was presented as a section within the Capital Structure Panel's testimony.

Q. In your testimony, do you refer to, or otherwise rely upon, any information obtained during the discovery phase of these proceedings?

A. Yes, I have referred to and relied upon responses to my interrogatories provided by the Companies. These responses are contained in Exhibit___(KXD-1). I will refer to them by the designation assigned by Staff, for example, DPS-100.

Q. Are you sponsoring any other exhibits?

A. Yes, I am sponsoring 34 exhibits as labeled below.

Exhibit Number	Description
Exhibit___(KXD-1)	Interrogatory Responses (IRs)

		from the Companies.
	Exhibit___(KXD-2)	KEDNY/KEDLI Capital Structure and Overall Rate of Return Matrix.
	Exhibit___(KXD-3)	National Grid plc's Corporate Structure.
	Exhibit___(KXD-4)	New York Ring-fencing Provisions.
	Exhibit___(KXD-5)	Moody's Investors Service (Moody's) Credit Opinion for KEDNY, Dated March 3, 2023.
	Exhibit___(KXD-6)	Moody's Credit Opinion for KEDLI, Dated February 20, 2023.
	Exhibit___(KXD-7)	S&P Global Ratings (S&P) Credit Opinion on KEDNY, Dated September 24, 2018.
	Exhibit___(KXD-8)	S&P's Credit Opinion on KEDLI, Dated September 24, 2018.
	Exhibit___(KXD-9)	S&P's Credit Opinion on KEDNY, Dated March 8, 2023.
	Exhibit___(KXD-10)	Duah's Calculation of Companies' witness Nowak's

		Proxy Groups' Average Common Equity Ratio from 2019 to 2022.
Exhibit___(KXD-11)		S&P's Credit Opinion on KEDLI, Dated February 10, 2023.
Exhibit___(KXD-12)		KEDNY/KEDLI Implied Credit Metrics.
Exhibit___(KXD-13)		S&P's Corporate Rating Methodology, Updated December 15, 2021.
Exhibit___(KXD-14)		Moody's Rating Methodology for Regulated Electric and Gas Utilities, Published June 23, 2017.
Exhibit___(KXD-15)		S&P and Moody's Definitions of Stable Outlook.
Exhibit___(KXD-16)		Article by James Pesando, entitled, "On Forecasting Long-Term Interest rates. Is the Success of the No-Change Prediction Surprising?"
Exhibit___(KXD-17)		Empirical Research Report: "Just How Bad Are

	Economists…" by Kevin Stephenson.
Exhibit___(KXD-18)	Debt Cost Calculation.
Exhibit___(KXD-19)	Duah's Proxy Group.
Exhibit___(KXD-20)	Summary of Duah's Cost of Equity.
Exhibit___(KXD-21)	Three-Month Stock Prices for Duah's Proxy Group.
Exhibit___(KXD-22)	Discounted Cash Flow Calculation for Duah's Proxy Group.
Exhibit___(KXD-23)	Energy Information Association Annual Energy Outlook, Dated March 16, 2023.
Exhibit___(KXD-24)	Relevant Excerpts from Bank of America Securities Quantitative Profiles, May 2023 to July 2023 Editions.
Exhibit___(KXD-25)	Duah's Capital Asset Pricing Model Results.
Exhibit___(KXD-26)	Illustration of Different Accounting Policy Choices on Operating Income.

Exhibit___(KXD-27)	Value Line Investment Survey, Quality Control Procedures.
Exhibit___(KXD-28)	Relevant Excerpts from the Book, "The Cost of Equity - A Practitioner's Guide (2010 Edition)" by David C. Parcell, Pages 88 and 89.
Exhibit___(KXD-29)	Article by Kent Baker et al. entitled, "Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice."
Exhibit___(KXD-30)	Empirical Research Report: "Whose Beta is Best?" by Ms. Diana Harrington.
Exhibit___(KXD-31)	Impact of Correcting the Flaws in Companies' witness Nowak's Return on common Equity Approach.
Exhibit___(KXD-32)	Regulatory Research Associates' Report on State Regulatory Evaluations, Dated February 24, 2023.
Exhibit___(KXD-33)	Regulatory Research

		Associates' Profile on New York State Regulatory Environment, Updated on March 24, 2023.
	Exhibit___(KXD-34)	Service Company Capital Structure and Overall Rate of Return for Asset Recovery Charge.

Rate of Return

Q. What overall, after-tax rate of return do you recommend for KEDNY for the Rate Year?

A. I recommend an overall, after-tax rate of return of 6.82 percent for KEDNY, as opposed to KEDNY's request of 7.09 percent.

Q. What overall after-tax rate of return do you recommend for KEDLI for the Rate Year?

A. I recommend an overall after-tax rate of return of 6.74 percent for KEDLI, as opposed to KEDLI's request of 7.05 percent.

Q. Please highlight the major differences between your rate of return recommendation and the overall rates of return requested by KEDNY and KEDLI.

A. The major difference between my recommended rate

of return and the Companies' requested rates of return is mainly attributable to my recommended return on equity, referred to as ROE, of 9.10 percent as opposed to the Companies' requested ROE of 9.80 percent. My recommended rates of return for both Companies are summarized in Exhibit___(KXD-2).

Q. What is the revenue requirement impact of your adjustment to KEDNY's requested rate of return?

A. The revenue requirement impact of my adjustment to KEDNY's rate of return is approximately $29.5 million on a pre-tax basis and approximately $19.85 million on an after-tax basis.

Q. What is the revenue requirement impact of your adjustment to KEDLI's requested rate of return?

A. The revenue requirement impact of my adjustment to KEDLI's rate of return is approximately $20.8 million on a pre-tax basis and approximately $14.6 million on an after-tax basis.

Q. What is 10 basis points in ROE and one percent in common equity worth for KEDNY?

A. For KEDNY, 10 basis points in ROE is worth approximately $5.2 million and one percent in common equity is worth approximately $5.6

million.

Q. What is 10 basis points in ROE and one percent in common equity worth for KEDLI?

A. For KEDLI, 10 basis points in ROE is worth approximately $3.24 million and one percent in common equity is worth about $3.56 million.

Fair Rate of Return

Q. Earlier you mentioned that the fair rate of return you recommend will be used to establish the revenue requirement for KEDNY and KEDLI. Please explain what you mean by revenue requirement.

A. In the context of Commission rate proceedings, the revenue requirement is the dollar amount required by a company to provide service during a rate year. It is the amount that will allow a utility to recover all of its reasonably expected operating costs, including taxes and depreciation. In addition, the revenue requirement includes a fair return that will allow the utility the opportunity to recover the cost of funds supplied to it by investors. The funds provided by these investors are needed for the company to finance its long-term utility

assets and working capital requirements, which in the rate-setting context is referred to as its rate base.

Q. Generally speaking, what is a fair rate of return for a regulated utility?

A. A fair rate of return for a regulated utility enables the company to provide safe and adequate service to its customers, while at the same time ensuring continuing support for the utility to have access to the capital markets for both its long-term debt and common equity securities, at terms that are reasonable given the company's level of risk. The fair rate of return, therefore, allows the utility to recover its prudently incurred cost of debt, while providing its common equity investors the opportunity to earn a return that is commensurate with the risk of their investment.

Q. How is a fair rate of return typically calculated?

A. The fair rate of return for a utility is calculated through a weighted average of the individual cost components of its expected capitalization during the rate year. The two

primary sources are long-term debt and common equity. Customer deposits, while typically a very small component, is generally reflected in a company's capitalization because it is a relatively permanent and stable source of capital employed by utilities. It should be noted that preferred stock is also sometimes a source of capital, although generally in much smaller proportions than either long-term debt or common equity. Since the Commission uses a fully forecasted rate year, it is also important that the rate year capitalization reflects the utility's projected capital requirements and that it is consistent with the goal of achieving the optimal cost of capital, particularly as it relates to the use of leverage.

Q. How are the cost rates of the individual cost components typically calculated?

A. The cost rates associated with a utility's long-term debt, preferred stock, and customer deposits are relatively straightforward to ascertain. Both the embedded cost of the long-term debt and preferred stock components can be readily calculated by examining their contractual terms,

e.g., the interest payments for the long-term debt and the preferred dividends for the preferred stock. The projected impact of the cost of any new or incremental long-term debt or preferred stock instruments, however, requires estimates using relevant market data. The cost rate for customer deposits is simply a matter of applying the interest rate that is currently prescribed by the Commission. The cost of common equity, however, is neither contractual nor prescribed by the Commission. Its calculation is complicated by the fact that all variables affecting a company's cost of equity cannot be directly observed, and instead, requires estimation and the opinion of analysts.

Q. Is the cost of common equity typically more expensive than the cost of long-term debt for a utility?

A. Yes. Even though both lenders and common equity investors supply the utility with the funds it needs to build and operate its systems, the common equity investors only earn a return after the payment of all other expenses. Because equity investors run the risk that their achieved

returns will not equal their expectations, the return required by a utility's equity investors is nearly always higher than that of the utility's long-term debt holders. Exceptions may exist during periods of disturbances in the market, such as during the recessionary period of 1980 to 1982, at which point the economy was beset with very high inflation and volatile interest rates. During that time, utility bond yields were at least as high as the equity returns the Commission allowed and far above the returns allowed by most state regulatory commissions.

Q. How can a utility's cost of common equity be measured?

A. The return requirements of a utility's common equity investors can be estimated through a cost of equity analysis. For almost three decades, the Commission has favored market-based methodologies such as the Discounted Cash Flow, or DCF, and the Capital Asset Pricing Model, or CAPM, to estimate the return required by equity investors.

Corporate Structure

Q. Please discuss the corporate structure at National Grid plc within which KEDNY and KEDLI operate.

A. As shown in Exhibit___(KXD-3), the Companies are wholly owned subsidiaries of National Grid USA, an intermediate public utility holding company with regulated subsidiaries engaged in the generation of electricity and the transmission and distribution of both natural gas and electricity. National Grid USA is a direct wholly owned subsidiary of National Grid North America Inc., which, in turn, is a wholly owned subsidiary of National Grid plc, the ultimate corporate parent of the Companies. National Grid plc is incorporated under the laws of England and Wales and is the holding company for a range of largely regulated businesses focusing on the ownership and operation of electricity and gas networks in the United Kingdom, and transmission and distribution utilities in the United States.

Capital Structure

Q. Do the Companies propose that their rates be set using their stand-alone capital structures or the

capital structures of either National Grid plc or National Grid USA?

A. The Companies propose to use their stand-alone capital structures reflecting a 48.0 percent common equity ratio for the Rate Year. It should be noted that the 48.0 percent common equity ratio is lower than KEDNY and KEDLI's actual stand-alone common equity ratios of 49.7 percent and 52.8 percent, respectively. However, the 48.0 percent common equity ratio is the same as what the Companies have been allowed in their two recent rate cases, specifically in 2016 and then again in 2019.

Q. Did the Companies provide any rationale for proposing their stand-alone capital structures for the Rate Year?

A. Yes, the Capital Structure Panel and cost of equity witness, Companies' witness Nowak, present two main reasons why it is appropriate to use the Companies' stand-alone capital structure for ratemaking purposes. First, the Companies believe that they implemented strong ring-fencing measures that have been recognized by the major rating agencies. Secondly, the Companies believe

that, for the four-year period ended 2021, utility operating subsidiaries in their proxy group have maintained an average common equity ratio of more than 6.0 percentage points above the 48.0 percent common equity ratio that the Companies have proposed in these rate proceedings.

Q. Can you explain the concept of ring-fencing?

A. Yes. Ring-fencing, with respect to a regulated public utility subsidiary, occurs when a subsidiary financially separates itself from its holding company/parent company to protect the subsidiary from financial instability or the bankruptcy of the parent. The ring-fencing measures intend to insulate assets in the subsidiary from the risks of the holding company/parent and the other subsidiaries in the holding company. In theory, if a subsidiary is ring-fenced, and the holding company or another subsidiary goes bankrupt, creditors would not be able to attach their claims to the assets of the ring-fenced subsidiary. In addition, a fully ring-fenced company should be insulated from its parent’s influence regarding the financial

decisions of the subsidiary. In theory, ring-fencing protects the assets of the subsidiary in the event of a bankruptcy of a parent or other affiliate. Moreover, ring-fencing may allow a subsidiary to be rated higher than the parent.

Q. Besides bankruptcy protection, what are the advantages of ring-fencing to a utility subsidiary?

A. If a utility subsidiary achieves a higher rating than its parent, there is a benefit of a lower cost of debt to the subsidiary and ultimately to ratepayers.

Q. What is your response to the Companies' claim that they have implemented strong ring-fencing measures which have been recognized by the major rating agencies?

A. I agree with the Companies that they have implemented strong ring-fencing measures that have been acknowledged by both Moody's and S&P.

Q. Please elaborate on the ring-fencing measures implemented by the Companies.

A. As a condition of accepting the Commission's approval of National Grid plc's acquisition of the Companies in 2006, both KEDNY and KEDLI

agreed to a series of ring-fencing provisions. These ring-fencing provisions, as well as those agreed to by other New York State utilities, are contained in Exhibit___(KXD-4). I will discuss the specific provisions of the ring-fencing measures later in my testimony.

Q. Please discuss Moody's March 3, 2023, Credit Opinion that referenced the Companies' ring-fencing measures.

A. Moody's March 3, 2023, Credit Opinion for KEDNY is contained in Exhibit___(KXD-5). The report mentions that KEDNY has strong ring-fencing provisions, which reduce the potential for debt to be pushed back down into the Company and increase its leverage. Additionally, the report highlights three credit-supportive provisions for KEDNY: (1) the "special preferred share" provision, which is designed to reduce the probability of bankruptcy in a distressed situation; (2) a requirement for an investment-grade credit rating; and (3) an explicit dividend restriction if KEDNY's debt/capitalization exceeds a 56 percent threshold. Moody's expressed a similar view about KEDLI's ring-

fencing measures in a February 20, 2023, report as shown in Exhibit___(KXD-6).

Q. Did Moody's reference any equity infusions from the parent company?

A. Yes, in its March 3, 2023, Credit Opinion for KEDNY, included as Exhibit___(KXD-5), Moody's specifically cited equity infusions from the parent company, highlighting their role in supporting KEDNY's financial profile.

Q. Comparatively, to what extent has S&P recognized that the ring-fencing mechanisms adequately insulate the Companies from their parent and affiliates?

A. In its 2018 credit opinions about KEDNY and KEDLI, issued separately on September 24, 2018, S&P stated that "…we assess [The Brooklyn Union Gas Co and KeySpan Gas East Corporation] as insulated, with existing insulation measures supporting a one-notch separation between the subsidiary and its parent."

Q. What factors did S&P list to support its assessment?

A. S&P listed four factors, stating, "[k]ey insulating measures include: [1] Prohibition of

dividend distributions if the company does not maintain investment-grade credit ratings with at least two of the three major rating agencies. [2] Prohibition of dividend increases and payments if the company's debt-to-capital ratio exceeds a certain amount. [3] A golden share whose vote is required to file the utility into voluntary bankruptcy. [4] Clear economic and strategic incentive from the parent to maintain [The Brooklyn Union Gas Company's] financial strength." These 2018 reports for KEDNY and KEDLI are included in Exhibit___(KXD-7) and Exhibit___(KXD-8), respectively.

Q. Do you have any observations about S&P's latest credit opinions about KEDNY and KEDLI?

A. Yes. S&P's latest credit report for KEDNY, dated March 8, 2023, reflects the benefit of a one-notch upgrade in its actual credit profile of "BBB+," compared to its stand-alone credit rating of "BBB." This one-notch upgrade is attributed to KEDNY's status as a core subsidiary of the parent companies, National Grid USA and National Grid plc. Despite KEDNY and KEDLI being designated by S&P as core subsidiaries of the

parent companies, KEDLI did not receive a similar one-notch upgrade in its credit profile from S&P. These credit opinions are included in Exhibit___(KXD-9) and Exhibit___(KXD-11).

Q. Based on this information, what is your conclusion regarding the strength of the Companies' ring-fencing mechanisms?

A. As stated earlier, it appears that the rating agencies still view KEDNY and KEDLI as having strong ring-fencing mechanisms, and thus I recommend the use of their respective stand-alone capital structures for ratemaking purposes.

Q. What is the Commission's policy on the appropriate capital structure to be used in ratemaking when a utility resides in a holding company organizational structure?

A. The Commission has expressed a preference for using the capital structure of the utility's parent company. The Commission elaborated on its position in its August 23, 2006, Order in Case 05-E-1222, stating:

"The established regulatory practice in New York in fully litigated rate proceedings, like this one, is to use the consolidated capital structure

of the holding parent company for ratemaking purposes… [w]e find no valid basis for excluding electric utility holding company structures from the time-honored and well-established regulatory practice. There is no rational basis for us to depart from the approach developed and used for consolidated telecommunication, natural gas, and water companies when addressing the substantially similar issues pertaining to consolidated electric company operations."

Q. Are there circumstances under which the Commission would allow the use of a stand-alone capitalization?

A. Yes. The Commission has indicated that it would consider using a stand-alone capitalization if a utility can demonstrate that it has adequately isolated itself from the risks of its affiliates.

Q. How can a utility demonstrate that it has isolated itself from the risk of its affiliates?

A. In its Abbreviated Order Authorizing Acquisition, issued September 9, 2008, and its Order Authorizing Acquisition Subject to Conditions, issued January 6, 2009, in Case 07-M-0906 where the Commission approved Iberdrola's acquisition

of Energy East, the Commission stated it would consider using a stand-alone equity ratio upon a showing that the rating agencies consider the utility operating subsidiary in question adequately insulated from the risks of the parent company's other operations. A company can establish such insulation by demonstrating the presence of significant ring-fencing measures.

Q. Does the existence of ring-fencing measures require the use of the subsidiaries' stand-alone capital structures in establishing the ratemaking capital structure?

A. Not necessarily. While the presence of ring-fencing measures provides the opportunity to use the utility's stand-alone capital structure, the financial policies of the utility must still be evaluated to identify whether a suitable financing mix, resulting from reasonable financing policies, is being deployed. The idea is to find the approximate ratios of debt and equity that lead to the lowest weighted cost of capital.

Q. Please describe what constitutes reasonable financial policies for a regulated utility.

A. A reasonable financial policy for a regulated utility provides an optimal mix of debt and equity financing. An optimal financing mix balances the interests of both ratepayers and investors by having the lowest cost of capital for an acceptable level of risk. A regulated utility, through its financial policies, should obtain a balanced mix of capital sources that will result in the most economical financing of its assets over the long run. For a utility company, a balanced capital structure will contain a sufficient proportion of common equity that will enable the utility to raise capital most economically. While common equity is costlier than debt, it should still be employed in sufficient amounts that will provide an optimal cost of capital.

Q. How can a utility obtain an optimal cost of capital?

A. A utility company can obtain an optimal cost of capital by balancing the cost and benefits of debt and equity financing to achieve the lowest long-run cost of capital.

Q. How do credit ratings influence a company’s cost

of capital?

A. Credit ratings play a significant role in determining a company's cost of capital. A higher credit rating typically allows a company to borrow at lower interest rates, thus reducing the cost of debt. Conversely, a lower credit rating might increase borrowing costs. Additionally, credit ratings might influence investor perceptions and a company's ability to raise equity. In essence, a favorable credit rating can contribute to a more optimal cost of capital.

Q. What credit ratings have Moody's and S&P assigned to KEDNY and KEDLI?

A. S&P assigns a credit rating of "BBB+" to KEDNY and KEDLI, both with a "Stable" outlook. KEDNY is currently rated "Baa2" by Moody's and KEDLI is currently rated "Baa1" by Moody's, again, both with a "Stable" outlook. These Moody's and S&P ratings are contained in Exhibit___(KXD-11), Exhibit___(KXD-5), Exhibit___(KXD-6), and Exhibit___(KXD-9).

Q. Have their "BBB+/BBB+" and "Baa2"/"Baa1" credit ratings provided the Companies with access to the

capital markets at reasonable rates?

A. Yes. In its August 2022 five-year fixed rate senior unsecured debt issuance of $400 million and ten-year fixed rate senior unsecured debt issuance of $450 million, KEDNY secured favorable rates of 4.632 percent and 4.866 percent, respectively. Similarly, on March 6, 2023, KEDLI issued $400 million of 30-year fixed-rate long-term debt at a reasonable interest rate of 5.994 percent.

Q. What did the Companies' witness say about the common equity ratio maintained by utility operating subsidiaries, within his proxy group, with similar business risk characteristics to KEDNY and KEDLI?

A. On pages 68 and 69 of his direct testimony, Companies' witness Nowak mentioned that he conducted an analysis utilizing data from SNL Financial, a business unit of S&P, for the years 2018 to 2021. His analysis concluded that the average common equity ratio of the utility subsidiaries for the holding companies in his proxy group was 54.13 percent. This ratio was about six basis points higher than their

requested 48.0 percent common equity ratio. Consequently, Companies' witness Nowak concluded that the Companies' requested 48.0 percent common equity ratio is reasonable and notably conservative.

Q. Do you have any concerns about Companies' witness Nowak's use of the 54.13 percent historical common equity ratio to support the reasonableness of the Companies' 48.0 percent common equity ratio proposal?

A. Yes. First, Companies' witness Nowak's use of the operating subsidiaries' average common equity ratio is inappropriate. Common equity investors, in his proxy group companies, base their return requirements on the holding company's use of leverage, along with its other risk attributes, as they are unable to directly invest in the subsidiaries. Second, the capital structures for utility subsidiaries of holding companies may not reflect either rational capitalization policies, or actual common equity employed, and therefore may not be suitable for supporting an operating subsidiary's rate of return. The subsidiary's common equity balance may not be financed by

common equity at the holding company level. Rather, some of the utility common equity balance may instead be proceeds from debt issued at the holding company level and classified on the utility subsidiary's books as common equity at the time the proceeds were invested in the utility subsidiary.

Q. Have you performed any analysis to ascertain the average common equity ratio of Companies' witness Nowak's proxy group?

A. Yes, I did. Using financial data from S&P Capital IQ, I found that the average common equity ratio of Companies' witness Nowak's proxy group for the years 2019 to 2022 was 45.34 percent as shown in Exhibit___(KXD-10). This is important because investors cannot invest in the equity of a wholly-owned subsidiary company. As such, equity investors look at, among other things, the overall capital structure of the consolidated parent company to determine their risk/reward outlook and make investment decisions.

Q. How was the proportion of common equity that KEDNY projected in the Rate Year forecasted?

A. Per KEDNY's Capital Structure Panel's Exhibit___(CSP-1), Schedule 1, the Rate Year proportion of common equity for KEDNY was forecasted by starting with the 2022 year-end capital structure and reflecting the impacts of forecasted sources and uses of funds from the beginning of 2023 through the Rate Year ending March 31, 2025. KEDNY's forecasted common equity ratio for the Rate Year is 49.7 percent. In developing its forecasted common equity ratio, KEDNY indicated that it plans to retain a portion of its earnings and use those earnings to address a portion of its funding needs, driven primarily by capital expenditures of approximately $1.012 billion in the Rate Year. Further, there is also a $50 million planned equity infusion from the ultimate parent, National Grid plc during the Rate Year. KEDNY plans to issue $500 million of long-term debt during the bridge period, from December 31, 2022, to March 31, 2024, and another $500 million of long-term debt during the Rate Year. For its customer deposits balance, KEDNY forecasts an amount of $13.236 million for the Rate Year, maintaining the same level as in the

prior two years.

Q. What is KEDLI's forecasted common equity ratio for the Rate Year?

A. KEDLI's Capital Structure Panel's Exhibit___(CSP-1), Schedule 2, indicates that KEDLI's forecasted common equity ratio for the Rate Year is 52.8 percent.

Q. How did KEDLI develop its forecasted 52.8 percent common equity ratio for the Rate Year?

A. In developing KEDLI's forecasted 52.8 percent common equity ratio for the Rate Year, the Company indicated that it plans to maintain most of its projected earnings and use them to address its funding needs, which will be driven primarily by capital expenditures of approximately $671 million during the Rate Year. Unlike KEDNY, which anticipates an equity infusion and plans to issue debt during the Rate Year, KEDLI does not expect an equity infusion but plans to issue $400 million of long-term debt during the Rate Year. For its customer deposits balance, KEDLI forecasts an amount of $14.13 million for the Rate Year, which aligns with the balance recorded in the two preceding fiscal years.

Q. Are the Companies requesting that their forecasted common equity ratios during the Rate Year be used for ratemaking purposes?

A. No, they are not. To mitigate their rate requests, the Companies propose to set rates based upon a capitalization with a 48.0 percent common equity ratio; the common equity ratio upon which their rates are currently set.

Q. Earlier, you mentioned that you support the use of the Companies' stand-alone capital structures for ratemaking purposes. Do you also agree with the 48.0 percent common equity ratio that the Companies have requested?

A. Yes, I recommend the use of a 48.0 percent common equity ratio for ratemaking purposes because I conclude that such a common equity ratio is reasonable.

Q. Why do you conclude that a common equity ratio of 48.0 percent is reasonable for ratemaking purposes for the Companies during the Rate Year?

A. Utilizing a common equity ratio of 48.0 percent for ratemaking purposes is reasonable as the Companies have adequate ring-fencing mechanisms in place that isolate the utility assets from the

ultimate parent. My determination that a 48.0 percent common equity ratio is reasonable is based on my pro forma financial analysis. This analysis demonstrates that a capitalization with a 48.0 percent common equity ratio, combined with my recommended ROE of 9.1 percent will ensure the Companies' financial integrity and provide them access to the capital markets at reasonable terms. This is further discussed in detail in the financial integrity section of my testimony.

Q. How did you determine the proportions of the respective capital components of the capital structure for KEDNY?

A. I based the capital components on the Staff Revenue Requirement Panel's determination of adjusted earnings base for KEDNY. First, I applied the 48.0 percent common equity ratio to the adjusted earnings base of approximately $7.3 billion to determine a common equity component of approximately $3.5 billion ($7.3 billion * 48 percent) in my recommended capital structure. Second, I included in my recommended capital structure KEDNY's projected customer deposits balance of $13.24 million for the Rate Year.

Third, after determining the common equity and customer deposits components, the residual of $3.78 billion, or approximately 51.82 percent of the adjusted earnings base, represents the projected long-term debt component for KEDNY.

Q. How did you determine the proportions of the respective capital components of the capital structure for KEDLI?

A. I based the capital components on the Staff Revenue Requirement Panel's determination of adjusted earnings base for KEDLI. First, I applied the 48.0 percent common equity ratio to the adjusted earnings base of $4.67 billion, to determine a common equity component of $2.24 billion ($4.67 billion * 48 percent) in my recommended capital structure. Second, I included in my recommended capital structure KEDLI's projected customer deposits balance of $14.13 million for the Rate Year. Third, after determining the common equity and customer deposits components, the residual of $2.42 billion, or approximately 51.70 percent of Staff Revenue Requirement Panel's adjusted earnings base, represents the projected long-term debt

component for KEDLI.

Financial Integrity Analysis

Q. Have you performed any analyses to substantiate your assertion that your recommended capital structure with a 48.0 percent common equity ratio, together with your recommended ROE, will be supportive of the Companies' financial integrity?

A. Yes, as shown in Exhibit___(KXD-12), my recommended common equity ratio and ROE when combined with other quantitative metrics and qualitative factors produce credit metrics that are generally consistent with the performance required by Moody's and S&P to maintain the Companies' current ratings of "Baa2"/"Baa1" and "BBB+"/"BBB+," respectively.

Q. Please discuss how you determined that the Companies' credit metrics are consistent with those ratings.

A. I based my analysis on the overall financial benchmark metrics used by Moody's and S&P in evaluating both the financial and business risks of regulated electric and gas utilities. I note that only S&P provides specific categories of

financial and business risks.

Q. Please discuss the concept of financial risk.

A. Financial risk is the additional risk a shareholder bears when a company uses debt in addition to equity financing. Companies that employ greater leverage would have higher financial risk than companies financed mostly or entirely by equity. Notably, S&P is the only credit rating agency that provides specific categories of financial risks. These categories include "Minimal," "Modest," "Intermediate," "Significant," "Aggressive," and "Highly Leveraged," and are based on seven different financial metric ratios. Reflected in Exhibit___(KXD-13) is the S&P Corporate Rating Methodology report originally published on November 19, 2013, and most recently updated on December 15, 2021. I incorporated these financial risk measures mentioned above in my credit metrics analysis in Exhibit___(KXD-12).

Q. Please elaborate on how S&P evaluates a company's risk.

A. As illustrated on pages 33 and 34 of the December 15, 2021, report contained in Exhibit___(KXD-13),

S&P calculates two core coverage ratios and six supplementary coverage ratios. Specifically, S&P examines the following financial risk ratios in assessing a firm's credit risk: (1) Funds from Operations (FFO)/debt; (2) Debt/EBITDA (Earnings Before Interest, Taxes, Depreciation, and Amortization); (3) Cash Flow from Operations (CFO)/debt; (4) Free Operating Cash Flow (FOCF)/debt; (5) Discretionary Cash Flow (DCF)/debt; (6) (FFO + interest)/cash interest (FFO cash interest cover); and (7) EBITDA/interest.

Q. What other factors did you use to analyze the Companies credit metrics?

A. Based on the methodology guidelines for the above metrics, I used S&P's medial volatility financial benchmark table reflecting the Companies' lower-risk, rate-regulated distribution operations. The medial volatility table contains a variety of financial ratios such as Debt-to-EBITDA and FFO-to-debt ratios. For each Company, I calculated the same financial ratios that are listed in the medial volatility table. Then, I compared the financial metrics that I calculated for each

Company with the benchmark financial ratios contained in the medial volatility table to determine the Company's financial risk category such as "Minimal," "Modest," "Intermediate," "Significant," "Aggressive," and "Highly Leveraged," as discussed earlier.

Q. What do you mean by "medial volatility" financial benchmark table?

A. Depending on S&P's assessment of a company's associated industry, country risk volatility, and cash flow volatility, S&P uses three volatility tables consisting of "standard volatility," "medial volatility," and "low volatility." S&P applies the "medial volatility" table to companies that have most of their operating cash flows from regulated activities with favorable regulatory conditions. This also applies to companies with about one-third or more of their consolidated operating cash flows coming from regulated utility activities with a supportive regulatory environment. I note that S&P applies the "medial volatility" table to all rated utilities in New York.

Q. In addition to the financial risk metrics in your

risk analysis of the Companies, did you consider other qualitative factors that were considered by S&P in its business risk analysis of the Companies?

A. Yes, I used an "Excellent" business risk profile in analyzing the credit metrics of KEDLI and a "Strong" business risk profile for KEDNY, which match the profiles utilized by S&P and Moody's. S&P's business risk profile analysis incorporates such factors as country risk, industry risk, regulatory environment, company competitive position, business and geographic diversification, and management strategy. Also, S&P's business risk profile categories include "Excellent," "Strong," "Satisfactory," "Fair," "Weak," and "Vulnerable." In determining the "BBB+" rating for the Companies, S&P has assigned "Strong" and "Excellent" business risk profiles to KEDNY and KEDLI, respectively.

Q. Please discuss S&P's rationale for assigning a "Strong" business risk profile to KEDNY.

A. As illustrated in my previously referenced Exhibit___(KXD-9), S&P assigns a "Strong" business risk profile to KEDNY, reflecting their

view that the Company's profitability and financial metrics have been declining in recent years.

Q. What is S&P's rationale for assigning an "Excellent" business risk profile to KEDLI?

A. As shown in my Exhibit___(KXD-11) referenced earlier, S&P's "Excellent" business risk assessment for KEDLI is based upon the Company's low operating risk regulated distribution operations, its somewhat small service territory that lacks geographic and operating diversity, a predominately residential and commercial customer base, and a generally constructive regulatory framework.

Q. What other factors did Moody's use to analyze the Companies?

A. My Exhibit___(KXD-14) contains the June 23, 2017, Moody's ratings methodology for regulated electric and gas utilities. Page four of the Moody's report contains the following financial strength metrics examined by Moody's: (1) Cash Flow from Operations (CFO), pre-Working Capital (WC) + Interest/Interest, with Moody's assigned weight of 7.5 percent; (2) CFO pre-WC/Debt, with

Moody's assigned weight of 15 percent; (3) CFO pre-WC- Dividends/Debt, with Moody's assigned weight of 10 percent; and (4) Debt/Total Capital, with Moody's assigned weight of 7.5 percent.

Q. Did you consider the Companies' business risk profile based on Moody's rating guidelines?

A. Yes, I did consider the business risk profile of the Companies based on Moody's rating guidelines for regulated electric and gas utilities as shown in my Exhibit___(KXD-14) referenced earlier. Moody's business risk profile analysis, which accounts for 60 percent of their overall indicative grid rating, incorporates such factors as: (1) Legislative and Judicial Underpinnings of the Regulatory Framework; (2) Timeliness of Recovery of Operating and Capital Costs; (3) Consistency and Predictability of Regulation; (4) Sufficiency of Rates and Returns; (5) Market Position; and (6) Generation and Fuel Diversity. Each of the six business risk factors is ranked as "Aaa," "Aa," "A," "Baa," "Ba," "B," and "Caa." Given that the Companies are rated by Moody's, I used Moody's qualitative scores for the Companies averaging "Baa1."

Q. How did you apply the 11 credits metrics consisting of the four financial metrics of Moody's and S&P's seven financial metrics discussed earlier?

A. As illustrated in my previously referenced Exhibit___(KXD-12), I applied these 11 total credit metrics using my recommended 48.0 percent common equity ratio and 9.1 percent ROE, along with the Staff Revenue Requirement Panel's and the Staff Gas Infrastructure and Operations Panel's recommendations for depreciation expense, current deferred income taxes, amortization of regulatory assets and liabilities, capital expenditures, and accumulated deferred income taxes. As Exhibit___(KXD-12) illustrates, KEDNY's and KEDLI's overall financial metrics, when combined with their respective "Strong" and "Excellent" business risk profiles, produce credit metrics consistent with S&P and Moody's credit ratings of "BBB-"/"Baa1" for KEDNY and "BBB+"/"A3" for KEDLI.

Q. For the purpose of comparison, did the Companies' Capital Structure Panel provide a projection of the Companies' financial metrics based on the

Companies' requested capital structure and cost rates?

A. Yes, the Companies' response to DPS-709, included in Exhibit___(KXD-1), provided the Rate Year credit metric projections. According to the Companies' projections, KEDNY's forecasted core credit metrics, except for the Debt-to-EBITDA ratio, are higher than its recent three-year historical average. Yet, KEDLI's forecasted core metrics are lower than its recent historical three-year average. As explained earlier, these credit metrics are based on the medial volatility and low-risk grid tables for S&P and Moody's reflecting the low business risk profiles of the Companies.

Q. How do your financial metrics and implied credit ratings compare with the financial metrics and ratings in the Companies' filing?

A. For KEDNY, my previously referenced Exhibit___(KXD-12) shows my implied Moody's/S&P credit ratings of "Baa1"/"BBB-" compared to the Company's "Baa1"/"BBB" provided in the response to DPS-709, which is included in Exhibit___(KXD-1). Similarly, for KEDLI, my implied Moody's/S&P

credit rating is “A3”/“BBB+” compared to “A3”/“A-” provided by the Companies in their response to DPS-709. One core coverage ratio used by S&P to assess the relative financial risk of regulated electric and gas utility companies is the leverage ratio, Debt to EBITDA. The ratio provides insight into a company’s ability to cover its fixed debt service (interest plus principal) obligations, as well as the variability of its earnings net of interest payments. A lower leverage ratio is considered more favorable as it indicates a stronger ability to meet financial obligations.

Q. How else does your financial metrics and implied credit ratings compare with the financial metrics and ratings in the Companies’ filing?

A. My fallout Debt-to-EBITDA coverage ratio is 4.97x and 4.83x for KEDNY and KEDLI, respectively, compared with the implied 4.77x and 4.52x for KEDNY and KEDLI, respectively, as provided by the Companies in response to DPS-709. Per S&P’s rating guidelines for regulated utilities, the appropriate Debt-to-EBITDA benchmark for a regulated electric and gas utility in the "BBB”

range with an excellent business risk profile is between 4.5x-5.5x, and 3.5x-4.5x for a "Strong" business risk profile.

Q. Is there any other S&P core metric that both you and the Companies used to analyze their financial metrics?

A. Yes, another S&P core metric used in both my and the Companies' analysis is the FFO/debt ratio. S&P uses the FFO/debt ratio to assess the financial health of regulated electric and gas utilities. The ratio provides insights into a company's ability to generate funds from its operations to service its debt. A higher FFO/debt ratio is considered more desirable as it is indicative of a stronger financial position and a greater ability to meet debt service requirements.

Q. How does your analysis of the Companies' FFO/debt ratio compare to the Companies' analysis of this ratio?

A. Under this metric, my capital structure recommendation along with the Staff Revenue Requirement Panel's and the Staff Gas Infrastructure and Operations Panel's

recommendations for depreciation expense, current deferred income taxes, amortization of regulatory assets and liabilities, capital expenditures, and accumulated deferred income taxes produces a FFO/debt ratio of 14.55 percent and 15.04 percent for KEDNY and KEDLI, respectively. Comparatively, in their response to DPS-709, the Companies produced an FFO/debt ratio of 15.19 percent and 15.81 percent for KEDNY and KEDLI, respectively.

Q. What do the S&P ratings guidelines say about the FFO/debt benchmark?

A. According to the S&P rating guidelines, the acceptable FFO/debt benchmarks reasonably expected from a regulated electric and gas utility in the "BBB" range with an "Excellent" business risk profile is between 9 percent to 13 percent and 13 percent to 23 percent for a "Strong" business risk profile. Yet, for the Rate Year, S&P is forecasting the FFO/debt ratio to be in the range of 13 percent to 14 percent for KEDNY and 13 percent to 15 percent for KEDLI.

Q. How does your analysis of the Companies' CFO Pre-Working Capital to debt ratio compare to the

Companies' analysis of this ratio?

A. As my previously referenced Exhibit___(KXD-12) illustrates, my fallout CFO Pre-Working Capital to debt ratio is 14.95 percent and 14.65 percent for KEDNY and KEDLI, respectively, compared with the implied 15.25 percent and 15.84 percent for KEDNY and KEDLI provided in the Companies' response to DPS-709.

Q. What do Moody's ratings guidelines say about the CFO Pre-Working Capital to debt ratio benchmark?

A. Per Moody's rating guidelines for regulated utilities, the appropriate benchmark for a regulated electric and gas utility in the "BBB" range with a low business risk profile is between 11 percent and 19 percent. As my previously referenced Exhibit___(KXD-12) illustrates, my fallout (CFO pre-WC/debt) metric is 14.95 percent and 14.65 percent for KEDNY and KEDLI, respectively, compared with the implied 15.25 percent and 15.84 percent for KEDNY and KEDLI provided in the Companies' response to DPS-709.

Q. What does S&P's most recent analysis say about the Companies' credit ratings outlook and the factors that could potentially lead to a rating

downgrade?

A. I reviewed the most recent credit analysis by S&P, as indicated in my previously referenced Exhibit___(KXD-9) and Exhibit___(KXD-11). According to S&P's analysis, there is a possibility of downgrading both KEDNY and KEDLI if their ultimate parent's consolidated FFO-to-debt ratio falls below 10 percent for a sustained period. Yet, it is important to note that S&P is forecasting a more favorable FFO-to-debt ratio of 13 percent to 14 percent for KEDNY and 13 percent to 15 percent for KEDLI. Additionally, S&P has assigned a "Stable" outlook to both KEDNY's and KEDLI's "BBB+" credit rating. A "Stable" outlook suggests that S&P does not anticipate any rating changes for the Companies in the intermediate term, which typically spans from six months to two years. Exhibit___(KXD-15) shows S&P and Moody's definitions of a "Stable" outlook.

Q. What is Moody's assessment of the Companies' outlook and the factors that could potentially lead to a rating downgrade?

A. In its latest credit opinion reports for the Companies as shown in my previously referenced

Exhibit___(KXD-5) and Exhibit___(KXD-6), Moody's assigned a "Stable" outlook for both KEDNY's and KEDLI's respective "Baa2" and "Baa1" credit ratings. According to Moody's, "[a] stable outlook indicates a low likelihood of a rating change over the medium term." This definition of "Stable" outlook is shown on page four of Exhibit___(KXD-15). Therefore, I believe Moody's does not view the Companies' current credit ratings as likely to change over the medium term. Yet, it is important to note that Moody's stated that KEDNY could be downgraded if the ultimate parent's consolidated CFO pre-WC/debt falls below 13 percent for a sustained period. Similarly, KEDLI could be downgraded if its own CFO pre-WC/debt is consistently below 15 percent.

Q. What is your opinion about the Companies' Capital Structure Panel's concerns regarding the potential negative impacts of further downgrades on the Companies access to the capital market and borrowing costs?

A. I recognize that further downgrades could negatively affect the Companies' access to the market and cost of borrowing. However, my

recommendations produce credit metrics that support the Companies' current credit ratings. This should be sufficient to provide them with access to the capital markets at reasonable rates during the Rate Year.

Cost of Long-Term Debt

Q. What weighted costs of debt are the Companies requesting in these rate proceedings?

A. KEDNY is requesting a weighted cost of debt of 4.60 percent and KEDLI is requesting a weighted cost of debt of 4.53 percent for the Rate Year.

Q. Please discuss the derivation of KEDNY's 4.60 percent weighted cost of debt for the Rate Year.

A. KEDNY's 4.60 percent weighted cost of debt reflects its 4.31 percent embedded cost of its outstanding long-term debt as of the historic test year, or the 12 months ending December 2022, along with the forecasted interest rates of 5.64 percent and 5.87 percent for a projected 10-year, $500 million debt issuance, and a projected 30-year, $500 million debt issuance, respectively. This totals $1.0 billion in projected senior unsecured debt issuances during the bridge period and the Rate Year.

Q. How did KEDNY determine the forecasted interest rates of 5.64 percent and 5.87 percent associated with the 10-year and 30-year projected senior unsecured debt issuances?

A. Per the Capital Structure Panel's Exhibit___(CSP-1), page two of Schedule one, KEDNY determined the 5.64 percent and 5.87 percent forecasted interest rates for the 10-year and 30-year projected debt issuances by adding the respective 3.594 percent and 3.519 percent of comparable US Treasury rates forecast as of March 10, 2023, to the Company's credit spreads of 205 basis points for the 10-year forecasted debt issuance and 235 basis points for the 30-year forecasted debt issuance. These credit spreads were derived from KEDLI's recent 10-year debt issuance that occurred on March 6, 2023, and the spreads were adjusted to account for the credit rating differential between KEDLI and KEDNY as well as the tenor of the 10-year and 30-year debt securities.

Q. Please discuss the derivation of KEDLI's 4.53 percent weighted cost of debt for the Rate Year.

A. KEDLI's 4.45 percent weighted cost of debt for

the Rate Year reflects its 4.45 percent embedded debt cost rate as of the historic test year, along with the forecasted interest rate of 5.80 percent for the 30-year projected debt issuance of $400 million.

Q. How did KEDLI determine the 5.80 percent forecasted interest rate for the projected 30-year debt issuance of $400 million?

A. KEDLI determined the 5.80 percent interest rate for the projected 30-year debt issuance by adding a comparable US Treasury rate forecast of 3.50 percent to the Company's 230 basis points credit spread. Again, this credit spread is derived from KEDLI's recent 10-year debt issuance that occurred on March 6, 2023, and the spread was adjusted to account for the term of the 30-year debt security.

Q. Did the Companies provide you with updated credit spreads and forecasted US Treasury yields?

A. Yes, in response to DPS-532, included in Exhibit___(KXD-1), the Companies provided updated credit spreads obtained from four investment banks as of August 10, 2023. For KEDNY, the credit spreads for the 10-year and 30-year

projected debt issuances increased from 205 and 235 basis points to 220 and 246 basis points, respectively. For KEDLI, the credit spread for the 30-year projected debt issuance increased from 230 basis points to 237 basis points.

Q. Do you have any concerns with the Companies' use of forecasted interest rates for its projected new debt issuances?

A. Yes, I do have concerns with the Companies' use of forecasted interest rates for its projected new debt issuances. My concerns are that such forecasts are not only poor predictors of the magnitude of the expected change in interest rates, but they are also not even reliable with respect to the direction of the change. Therefore, I recommend the use of the current Treasury yields plus the Companies' credit spread to estimate the interest rates for their projected new debt issuances because current interest rates are the best predictors of future interest rates. Moreover, this approach is more reliable in predicting future interest rates particularly for a one-year rate plan as in these rate proceedings.

Q. Are you aware of any academic publications supporting your assertion that current interest rates are the best predictors of future interest rates?

A. Yes, I am aware of two publications. The relative predictive power of current interest rates versus forecasted interest rates was discussed in a study entitled, "On Forecasting Long-Term Interest Rates: Is the Success of the No-Change Prediction Surprising?" by Dr. James E. Pesando in the Journal of Finance, September 1980, included as Exhibit___(KXD-16). In the study, the author argued that the best estimate of future long-term interest rates is "no-change," meaning the current rates of debt instruments. The second publication is entitled "Just How Bad Are Economists At Predicting Interest Rates? (And What Are The Implications For Investors)" authored by Kevin Stephenson. The author examined whether economists could predict interest rates and he concluded through his empirical research that the "answer is emphatically no, regardless of the measure used." This is shown as Exhibit___(KXD-17).

Q. Is there any other reason why you are recommending the use of current US Treasury yields plus the Companies' respective credit spread to estimate the interest rates for the Companies' planned debt issuances?

A. Yes, because the Commission considers current interest rates to be the best predictor of where interest rates will be in a given rate year going forward. Specifically, in its January 24, 2011, Order in Case 10-E-0050, Proceeding on Motion of the Commission as to the Rates, Charges, Rules, and Regulations of Niagara Mohawk Power Corporation for Electric Service, the Commission supported this method. In the January 24, 2011, Order, the Commission stated that "[i]n as much as we are making a one-year rate determination, the current interest rates are reasonably representative for the immediate future."

Q. What would be the interest rates for KEDNY's new debt issuances using your recommended approach?

A. Using my recommended approach of utilizing the most recent 10-year and 30-year monthly average of the US Treasury yields of 3.90 percent and 3.96 percent for July 2023, respectively, along

with the Company's respective 220 and 246 basis point updated credit spreads, I estimate an interest rate of 6.10 percent for KEDNY's forecasted 10-year and 6.42 percent for KEDNY's 30-year forecasted debt issuances. I will explain later why I used the Companies' proposed credit spreads.

Q. What would be the impact on KEDNY's overall weighted average cost of debt using your 6.10 percent and 6.42 percent interest rates for the forecasted debt issuances?

A. Using my estimated 6.10 percent and 6.42 percent interest rates for KEDNY's forecasted debt issuances would result in a 4.70 percent overall cost of debt, as illustrated in Exhibit___(KXD-18). Therefore, I recommend a 4.70 percent overall weighted cost of debt for KEDNY.

Q. Why did you use the Companies' proposed credit spreads?

A. I utilized the Companies' proposed credit spreads because they are in line with the credit spreads of their existing debt instruments that are actively traded in the market, indicating that their proposed spreads are reasonable.

Q. Would you explain how you utilized the Companies' credit spreads in your analysis?

A. Yes. As detailed on pages three and four of Exhibit___(KXD-18), I analyzed the credit spreads of the Companies' existing debt instruments that are actively traded in the market. By comparing the yields on these bonds with the yield on risk-free government bonds of similar maturity, I determined the credit spreads. I further validated my findings using S&P Capital IQ's spread-to-worst data. Spread-to-worst data helps investors understand the worst-case yield scenario for a bond. It is the least favorable or the lowest yield that bond investors can expect, given all potential call or prepayment scenarios, excluding default.

Q. What were the results of your analysis for KEDNY?

A. For KEDNY, the 30-year debt instruments issued between 2016 and 2019, with tenors as of July 2023, ranging between 23 and 26 years, have credit spreads ranging between 225 and 245 basis points above Treasury benchmarks. S&P Capital IQ's spread-to-worst data for these debt instruments ranges between 225 and 243 basis

points. Additionally, KEDNY's 10-year debt instruments issued between 2019 and 2022, with tenors ranging between six and nine years, have credit spreads ranging between 181 and 190 basis points, with spread-to-worst data values between 161 and 191 basis points. I also performed a similar credit spread analysis for KEDLI.

Q. What were the results of your analysis for KEDLI?

A. For KEDLI, the 30-year debt instruments issued between 2011 and 2022, with tenors as of July 2023 ranging between 18 and 28 years, show credit spreads between 207 and 234 basis points, with S&P Capital IQ's spread-to-worst data values between 206 and 214 basis points. KEDLI's 10-year debt issued in March 2023, has a credit spread of 206 basis points, as of July 2023. These credit spread figures for both KEDNY and KEDLI compare favorably with those proposed by the Companies. It is important to note that my credit spread figures do not account for differences in tenor, as of July 31, 2023, of existing debt and the maturity of planned debt issuances. Adjusting for these factors would yield higher results.

Q. What would be the interest rate for KEDLI's planned debt issuance using your recommended approach?

A. Using my recommended approach of utilizing the most recent monthly average, as of July 2023, of the 30-year Treasury yield of 3.96 percent, along with KEDLI's 237 basis points updated credit spread, the interest rate would be 6.33 percent for KEDLI's 30-year planned debt issuance, as illustrated in Exhibit___(KXD-18).

Q. What would be the impact on KEDLI's overall weighted average cost of debt using your 6.33 percent interest rate for the planned 30-year debt issuance?

A. Using my estimated 6.33 percent interest rate for KEDLI's planned debt issuance, would result in a weighted average cost of debt of 4.56 percent, as illustrated in Exhibit___(KXD-1). Therefore, I recommend a 4.56 percent overall weighted cost of debt for KEDLI.

Q. Do you have any recommendation regarding the Companies' weighted cost of debt rates?

A. Yes, I do. First, I recommend that the Companies' overall weighted cost of debt rates be

updated prior to the Commission decision. This update should include the actual interest rates obtained by the Companies for their projected new debt issuances during the bridge period. Additionally, I recommend updating the interest rates for the Companies' planned debt issuances for the Rate Year, taking into account the Companies prevailing credit spreads and Treasury yields prior to the Commission decision, and ensuring that these updated interest rates are accurately reflected in the Companies' respective overall cost of debt for the Rate Year.

Q. Do you agree with the Companies' proposal to implement a true-up mechanism for the interest rates associated with their new fixed-rate debt issuances for the Rate Year?

A. I do not agree with the Companies' proposal to implement a true-up mechanism for the interest rates associated with KEDNY's $1.0 billion and KEDLI's $400 million debt issuances for the Rate Year. Given the current stability in the capital markets, the Companies should be able to access the markets with reasonable certainty during the Rate Year without the need for a true-up.

Cost of Common Equity

Summary

Q. What methodology did you use to determine your recommended 9.1 percent ROE?

A. I estimated the cost of equity for a proxy group of 31 regulated utility companies classified by Value Line as "electric utility" and "natural gas utility" holding companies using a DCF analysis, weighted two-thirds, and the average of two CAPM analyses, weighted one-third, which resulted in a 9.07 percent ROE, rounded up to my recommended 9.1 percent.

Q. Please explain why you specifically recommend that the DCF methodology be given a two-thirds weighting and the CAPM result be accorded a one-third weighting.

A. For more than 25 years, the DCF has been the principal equity costing methodology in New York. In fact, the Commission has consistently issued cost of equity determinations with the same two-thirds DCF and one-third CAPM weightings since 1994. During this time, Staff Financial Analysts' testimonies addressing ROE have consistently noted the numerous reasons why the

DCF has been and should continue to be the preferred methodology. For example, Staff Financial Analysts have advocated that the CAPM be weighted less than the DCF in the overall ROE calculation because the DCF has proven to be less volatile over time and is more widely used throughout the country in utility rate case settings. Estimating the cost of equity requires using imperfect methodologies. Of all the approaches available, however, the DCF and the CAPM are by far the least flawed, and between the two, the DCF is superior.

Q. In which cases have Staff Financial Analysts advocated that the CAPM be weighted less than the DCF?

A. Staff Financial Analysts have advocated that the CAPM be weighted less than the DCF in numerous cases. These cases include Staff's Finance Panel testimony, on pages 50-51, filed on September 7, 2007, in Case 07-E-0523 regarding Consolidated Edison Company of New York, Inc. Also, the Staff Finance Panel testimony, on pages 70-71, filed on July 14, 2010, in Case 10-E-0050 involving National Grid, advocated for a similar approach.

Q. What has the Commission said about the DCF methodology?

A. The Commission has noted the relative strengths of the DCF methodology in many of its previous rate orders. For example, in Case 06-E-1433, on page 14 of the Order Setting Permanent Rates, Reconciling Overpayments During Temporary Rate Period, and Establishing Disposition of Property Tax Refunds, issued on October 18, 2007, the Commission stated that the two-stage DCF "method offers the significant benefit of reliance on readily available, objective data to measure an indicator of real importance to investors."

Staff Proxy Group

Q. Why did you use a proxy group to estimate the cost of equity for the Companies?

A. The use of a proxy group to determine the cost of equity is necessary for several reasons. First, the Companies' common stocks are not publicly traded; thus, direct DCF and CAPM analyses of the Companies are not possible. Second, DCF analyses for individual companies rely on equity analysts' estimates of growth, and those estimates could be biased or potentially inaccurate. Similarly,

beta determinations used in the CAPM methodology are based on historical observations that, due to circumstances such as corporate restructurings or major industry transformations, may not be representative of the level of earnings volatility expected in the future.

Q. What is the benefit of utilizing a large proxy group in your analysis?

A. By employing a sufficiently large proxy group of similarly situated companies in my analysis, I can minimize the undesirable effects of bias, both upward and downward, or inaccurate estimates for any one company.

Q. What are the most important considerations for selecting a proxy group?

A. First, it is important to determine the specific industry classification of the subject company to identify its true peers. Second, once the appropriate group of peer companies is established, careful consideration must be given to determining the appropriate screening criteria to establish a group of companies that is both sufficiently large and of similar risk to the company in question.

Q. Why is it important to balance these two considerations when selecting a proxy group?

A. A careful balance must be struck between these two potentially conflicting goals. While the objective is to select a group of companies whose business and financial risks closely match those of the subject company, it is also important that the proxy group be large enough to provide sufficient confidence in the results. The greater the number of suitable companies that can be found, the less sensitive the overall cost of equity estimate will be to the fluctuations or irregularities in the data related to any one company.

Q. Please describe how you selected your proxy group.

A. I selected a group of 31 utility holding companies from a universe of 47 publicly traded utility holding companies that Value Line categorized as “electric utilities” and “natural gas utilities.” In this Value Line universe, Value Line categorized 38 holding companies as “electric utilities” and nine holding companies as “natural gas utilities.” Due to the robust

size of my proxy group, I am confident that it will produce a reliable estimate of the Companies' cost of equity.

Q. What criteria did you use to select companies for your proxy group?

A. I carefully selected companies that face risks similar to those faced by the Companies. The list of companies that I used in my analysis, along with the current credit ratings assigned to each by S&P and Moody's, accompanied by the percentage of revenues that are derived from regulated activities for each company during the 12 months ended December 31, 2022, can be found in Exhibit___(KXD-19).

Q. What screening criteria did you apply to your proxy group?

A. Beginning with the 47 publicly traded holding companies that Value Line categorizes as "electric utilities" and "natural gas utilities," I applied four selection criteria to match the risks of the Companies. The four criteria require that each company: (1) must currently maintain an investment-grade credit rating from Moody's and S&P; (2) must have regulated utility

revenues of at least 70 percent of its total revenues, as determined by each company's 2022 form 10-K that was filed with the Securities and Exchange Commission; (3) must currently pay dividends; and (4) must not be engaged in any major mergers, acquisitions, or other transformational activities.

Q. Why did you limit your proxy group to those companies that have an investment grade credit rating?

A. A credit rating represents the creditworthiness of a company and the risk that it may not be able to repay its loans. Fixed income and equity investors seek to measure the level of risk associated with the companies in which they invest, which is reflected in the credit rating. Most of the parent holding companies in the Value Line universe of utility holding companies hold an investment-grade credit rating.

Q. Why did you limit your proxy group to those companies that derive at least 70 percent of their total revenues from regulated operations?

A. The Companies, as transmission and distribution utilities, derive all their revenues from

regulated utility operations. To ensure greater comparability with the Companies, I eliminated companies that derive a significant portion of their total revenues from non-regulated businesses, as such activities generally have materially higher business risk than a utility operating company. Ideally, the proxy group would consist entirely of utility operating companies like KEDNY and KEDLI. Yet, this is not a reasonable expectation given the diversification into competitive businesses that utility companies have generally pursued over time. The 70 percent threshold I use achieves a reasonable balance that enables me to select a proxy group that is large enough to provide a valid comparison to the Companies, while excluding companies that would distort the group's risk profile because they rely too heavily on unregulated revenues.

Q. Why do you exclude companies that are currently not paying dividends?

A. A company's current stock price and recent dividends are two core components necessary to evaluate the present value of a company's

estimated return on equity in the DCF model. Without these key elements, estimating the ROE with the DCF model would not be feasible.

Q. Why do you exclude companies that are currently party to, or have recently completed, substantial merger or acquisition-related activity?

A. Transformational financial transactions, such as large mergers and acquisitions, can distort the common stock prices of the companies involved, thereby undermining the reliability of the cost of equity estimates. For example, recently completed mergers can lead to a new entity that does not have a sufficient beta or dividend history. Also, analysts may not have assessed or incorporated the full impact of the merger on future dividends and earnings.

Q. Which companies were excluded from your proxy group and why?

A. I excluded companies that did not provide a fair comparison with KEDNY and KEDLI. First, I excluded three companies due to their lack of investment grade credit ratings: FirstEnergy Corp., PG&E Corp, and Hawaiian Electric Industries Inc. Although S&P has assigned a

"BBB-" rating to FirstEnergy, which is an investment grade, Moody's has assigned it a "Ba" rating, which is not investment grade. Because "BBB-" is the lowest investment grade rating, I think that the higher risk implied by this split rating would not be appropriate for inclusion in the proxy group. In quick succession, Hawaiian Electric Industries, Inc. faced two downgrades by S&P in August 2023 in the wake of the catastrophic wildfires in the company's service territory in Maui. Initially, on August 15th, S&P lowered its rating from "BBB-" to "BB-", followed by a further downgrade to "B-" just nine days later, on August 24th. Similarly, on August 18, Moody's assigned a Corporate Family Rating of "B1" to Hawaiian Electric Industries, Inc. and further placed it under review for further downgrade.

Q. What other companies did you exclude from the proxy group?

A. Next, I excluded three companies, Chesapeake Utilities Corp., UGI Corp., and New Jersey Resources Corp, because they are not rated by Moody's and S&P. Also, I excluded four

companies, DTE Energy Company, Otter Tail Corporation, PNM Resource Inc., and Southwest Gas Holdings, Inc., because they do not derive at least 70 percent of their total revenues from regulated operations. Further, I eliminated Fortis, Inc. because it reports its financial results in a foreign currency, which would introduce foreign exchange-rate risk into my proxy group.

Q. What other companies did you exclude from the proxy group?

A. I eliminated MGE Energy Inc. and Unitil Corporation because they lack Value Line dividend forecasts and sufficient financial data needed to calculate the DCF result. Also, I eliminated Exelon Corp. because it lacks a beta from Value Line, which is needed to calculate the CAPM results. Finally, NiSource and Avangrid were excluded due to their separate acquisition-related transactions. Specifically, in a June 20, 2023, press release, Avangrid Inc. and PNM Resources, Inc. announced that the two companies have mutually agreed to amend and extend the agreement for their previously announced merger,

which carries an enterprise value of $8.3 billion, until through December 31, 2023. Similarly, on the same day, NiSource announced its plans to divest a 19.9 percent equity interest in Northern Indiana Public Service Co. LLC to an affiliate of Blackstone Infrastructure Partners for $2.15 billion.

Q. What are the results based on your selection criteria?

A. In total, I excluded 16 of the 47 electric and natural gas utility companies covered by Value Line. The remaining 31 electric and natural gas utilities that comprise my proxy group are illustrated in Exhibit___(KXD-19).

Q. Why is your proxy group appropriate for determining the Companies' cost of equity?

A. My proxy group is appropriate for determining the Companies' cost of equity because all the parent utility holding companies in the proxy group serve retail customers, have investment grade senior unsecured debt ratings, receive a minimum of 70 percent of total revenues from regulated operations, and are not involved in any transformational proceedings, such as a merger or

acquisition.

Q. Has the Commission employed Staff Financial Analysts' proxy group in its cost of equity determinations in previous rate orders?

A. Yes, the Commission has relied on Staff Financial Analysts' proxy group in ROE determinations in previous rate orders. For example, in its April 20, 2017, Order Establishing Rates For Gas Service in Case 16-G-0257, involving National Fuel Gas Distribution Corp., the Commission found the composition of the Staff Financial Analysts' proxy group to be superior and employed it to derive its ROE determination. Further, more recently in its May 19, 2021, Order Establishing Rates and Rate Plan issued in Case 20-G-0101, which concerned Corning Natural Distribution Corporation, the Commission once again relied on the Staff Financial Analysts' proxy group analysis due to its reasonableness. On page 46 of the May 19, 2021, Order, the Commission stated, "This result is the product of a rational methodology that creates predictability and certainty and balances all interests involved."

Q. Please summarize the characteristics of your

proxy group concerning the average credit rating and percentage of regulated revenues.

A. As illustrated on page one of my Exhibit___(KXD-19), the average S&P rating of my proxy group is "BBB+" and the average Moody's rating is "Baa1." On average, the companies in my proxy group receive approximately 93 percent of their revenues from regulated utility operations.

Discounted Cash Flow Methodology

Q. Please explain the basic theory underlying the DCF methodology.

A. The DCF approach can be applied to any investment instrument that has an intrinsic value. The DCF approach, as it relates to common stock, recognizes that companies create value for their stockholders by using their earnings in several ways, the most important of which, by far, is through the payment of cash dividends. Alternatively, earnings that are retained by companies can be used to create value by investing in capital projects designed to increase future profits. The retained earnings can also create value by retiring debt, which reduces interest expense, thereby resulting in

greater cash flow available to stockholders, and by buying back some of the company's common stock, which increases future earnings on a per-share basis.

Q. Under the DCF methodology, how is a company's common stock valued against future dividends?

A. It is important to note that, while earnings drive companies' dividend payout policies, the value of the companies' common stock is always equal to the present value of all future dividends. This is because the earnings that are retained will only have value to stockholders when they are paid out to stockholders in the future. Underlying this principle is the strong assumption in capital market theory that companies earn the same return on retained earnings as the market demands on their common stock. The DCF theory assures us that stocks only have value because of the cash flows that current investors receive or the anticipation of cash flows that future investors hope to receive.

Q. Under the DCF methodology, how is future cash valued?

A. Fundamental to the DCF methodology is the notion

that cash in the future is not worth as much as cash today. This is due to reasons such as the preference of individuals to consume today rather than in the future, and the effects of inflation and productivity on expected future cash flows. Accordingly, the DCF discounts the future expected cash flows according to investors' return requirements.

Q. Why do you place primary reliance on the results of the DCF methodology?

A. The main reason the DCF methodology continues to be the preferred approach for determining a company's cost of equity is that investors' immediate return requirements, as observed in current stock prices and recent dividends, are readily quantifiable. The other principal methodology, the CAPM, only relies tangentially on direct observations of actual utility investor behavior using utility beta values.

Q. What is the challenge in utilizing the DCF methodology?

A. The primary challenge in applying the DCF is determining the rate of growth in future dividends that investors expect. Given the

relatively mature and stable nature of the utility industry, estimates can be derived with a reasonable degree of certitude. Also, rational utility investors expect the growth in future dividends to generally track the changes in output, or growth in the overall economy, as measured by growth in the nominal Gross Domestic Product, or GDP. Moreover, just as nominal GDP growth incorporates gains achieved through the application of new technologies, otherwise known as productivity, and the effects of changes in price levels, the investors' growth expectations will also reflect assumptions regarding productivity and the rate of inflation. Consequently, when the DCF is utilized with the application of well-reasoned growth rate estimates, such as those used in my approach, the intuitiveness of the DCF methodology is clear.

Q. Why has the Commission regularly found the DCF methodology to be a reliable methodology for estimating a regulated utility's cost of equity?

A. The intuitiveness of the DCF methodology is the primary reason that the Commission has regularly found this methodology to be the best tool for

estimating the cost of equity for a regulated utility. In its April 24, 2009, Order Setting Electric Rates in Case 08-E-0539 regarding Consolidated Edison Company Of New York, Inc., the Commission stated that "…the DCF relies on readily available data to make objective estimates of investors' return requirements. While the DCF has one input of primary controversy (growth), two CAPM inputs (beta and the market risk premium) are dependent on estimates which are contested and volatile."

Q. Please describe the DCF methodology you employed to determine the cost of equity for the Companies.

A. I employed a two-stage DCF methodology to determine the cost of equity for the Companies. The two-stage DCF methodology is a "dividend discount" model that is consistent with the financial theory that the value of a company's stock is equivalent to its discounted future cash flows. My dividend discount model forecasts those cash flows into the future and discounts them back to their present value. This model also explicitly recognizes that a company's

short-term growth expectations do not necessarily equal long-term expectations; hence, my use of a two-stage DCF model.

Q. Why is a two-stage DCF Methodology preferable to a single-stage DCF Model?

A. Such a model is preferred, especially when growth rates in the short-term and long-term might reasonably be expected to diverge, thus making it superior to a single-stage DCF model, which assumes constant growth. For each company in my proxy group, a three-month average stock price was calculated by averaging the high and low prices in each month for the three months ended July 2023, as shown in Exhibit___(KXD-21). The model also contains Value Line data for earnings per share, or EPS, dividends per share, book value per share, and the forecasted amount of outstanding common stock for each company.

Q. How do you use this methodology to determine the appropriate rate of return?

A. This data is used to estimate the current price that investors are willing to pay for the future stream of dividends, represented in this instance by the average stock price taken over the three

months ended July 2023. This is essentially the present value of those expected dividends. By calculating the discount rate required to turn the string of expected dividend payments into the current stock prices, I identify the rates of return that investors expect for each company. The calculation of the DCF for my proxy group is illustrated on pages one and two of Exhibit___(KXD-22).

Q. How are dividends projected to change over time?

A. My analysis maintains the same approach that Staff Financial Analysts have used for many years. In the first stage of my DCF model, from 2023 through 2027, I rely on Value Line analysts' forecasted dividends for each company in my proxy group. In the second stage, for the years 2028 and beyond, I calculate a "sustainable growth" rate for each company within the proxy group. I find that using analyst forecasts for dividend growth in the near term is reasonable, but analysts generally do not publish dividend forecasts for longer than a few years into the future. So, unless one makes the unreasonable assumption that near-term estimates of growth

will continue indefinitely into the future, it is essentially impossible to employ analyst forecasts of dividend growth over the long term. Instead, I calculate a sustainable growth rate for the long term. I then calculate the cost of equity by solving for the discount rate necessary to set the net present value of the 200-year dividend stream for each company equal to its current stock price. The calculation of the proxy group's dividend growth as measured by the DCF methodology is displayed on page two of Exhibit___(KXD-22).

Q. Explain both the concept and the derivation of the "sustainable growth" rate for the DCF.

A. Sustainable dividend growth rates depend on a firm's expected internal and external growth rates. Long-term internal growth will be achieved based upon a company's earned return on common equity and on the portion of earnings retained within the business. That is, internal growth, also known as retention growth, is achieved based upon the growth in a company's retained earnings.

Q. Would you explain the formula that you use to

calculate retention growth?

A. In formulaic form, retention growth is measured by the following formula:

Retention growth = "b"*"r"

Where:

"b" = the fraction of earnings retained in the company, and "r" = the expected rate of return on common equity. For example, if a firm retains 40 percent of its earnings and earns 10 percent on equity, then its retention growth rate will be 4.0 percent (40 percent * 10 percent). Thus, the growth rate is determined by the increase in the equity retained in the business and the expected return on that equity. My analysis calculates median and average retention growth rates of 3.95 percent and 4.19 percent, respectively for the proxy group.

Q. How is external growth measured in your DCF methodology?

A. External growth can be achieved if a company issues common equity above its book value. This is commonly referred to as the SV factor, where "S" is the increase in common shares outstanding that have been sold and "V" is the per share

premium or discount to the book value on the shares sold. Typically, Staff Financial Analysts evaluate the SV factor using the formula: (MBR-1) * growth in common shares, where: (MBR) is the market-to-book ratio. Thus, the formula for the sustainable growth rate is given by g = ("b" * "r") + (SV), where SV is also given by (MBR-1)* growth in common stock. In my analysis, the median and average external growth rates for the proxy group are 0.33 percent and 0.45 percent, respectively. The sum of the internal and external growth rate forecasts is used to establish the sustainable growth rate.

Q. What is the sustainable growth rate of your proxy group?

A. The mean sustainable growth rate is 4.63 percent. This is illustrated on page two of my Exhibit__(KXD-22).

Q. Did you check the reasonableness of your proxy group's sustainable growth rate with any macroeconomic indicators?

A. Yes. I compared the sustainable growth rate result of my proxy group with two reputable publications that are well-known for reporting

macroeconomic forecasts. These publications are the Blue Chip Economic Indicators report, dated March 10, 2023, and The U.S. Energy Information Administration's, or EIA's, 2023 Annual Energy Outlook report, dated March 16, 2023.

Q. What are the growth rate estimates in the Blue Chip Economic Indicators report?

A. The consensus long-range estimate for nominal GDP growth is 4.10 percent for the 2025 to 2029 period and 3.9 percent for the more distant period forecast, 2030 to 2034.

Q. What does the average sustainable growth rate for your proxy group indicate?

A. My proxy group's 4.63 percent average sustainable growth rate is comparable to the consensus estimate growth rate in the overall economy, if not slightly optimistic on utility growth prospects, considering one could reasonably expect a mature industry such as the utility sector to grow at a slower rate than the overall economy. It should be noted that the 4.10 percent nominal GDP growth rate estimate for the 2025 to 2029 time period is comprised of two components: real GDP growth of 1.9 percent and an

inflation rate of 2.20 percent. The long-run projections generally show annual real GDP growth decreasing from a rate of 1.9 percent in 2025 to a rate of 1.80 percent in 2028, while inflation is forecasted to decline from 2.2 percent in 2025 to a rate of 2.1 percent in 2029 and beyond into the long run. This comparison is appropriate because the nominal GDP rate reflects assumptions about future inflation, in addition to the real growth expected in the economy because of productivity gains.

Q. What does the data indicate for future dividends?

A. It would not be unreasonable for investors in the market as a whole to expect their future dividends to keep pace with overall inflation, as well as reflect productivity gains similar to those expected for the economy as a whole. Likewise, for investors in a mature sector of the economy with slower-than-average growth prospects, such as the utility industry, it is reasonable to expect future dividend growth to be roughly equivalent to that of the overall economy.

Q. Please discuss the growth rate estimates in the

EIA's March 2023 Annual Energy Outlook report and how they compare to your sustainable growth rate.

A. I compared my 4.63 percent average sustainable growth rate to EIA's 2023 Annual Energy Outlook report, as shown on page four of Exhibit___(KXD-23). It is important to note that the EIA's report provides real GDP forecasts, as opposed to the Blue Chip Economic Indicators report which provides nominal GDP forecasts. EIA's base case scenario estimates that the real GDP will experience 1.90 percent annual growth from 2022 to 2050, whereas EIA's high growth scenario estimates a 2.30 percent annual growth in real GDP for the same period. Finally, EIA's low growth scenario estimates real GDP growth of 1.40 percent annually.

Q. How does EIA's estimate change when inflation is factored in?

A. Since EIA's estimates did not account for inflation, I added EIA's estimate of real GDP growth to the inflation forecast of 2.20 percent published by Blue Chip Economic Indicators, which yielded nominal GDP growth estimates in the range of 3.60 percent to 4.50 percent. Similar to the

estimates published in the Blue Chip Economic Indicators, EIA's estimates are comparable to my 4.63 percent average sustainable growth rate. In this case, my estimate is on the higher end of EIA's estimated range.

Q. What cost of equity does your DCF methodology produce for your proxy group?

A. The average DCF ROE result of my proxy group is 8.52 percent, as illustrated on page two of Exhibit___(KXD-22).

Q. Please explain why you recommend using the average DCF result, rather than any of the individual company results within the proxy group.

A. While most of the individual company results appear reasonable, I would not recommend a cost of equity based on any one of the individual results because of the potential for biased or inaccurate financial data estimates influencing the results. The simple fact remains that financial data forecasts, even in the relatively stable utility industry, can be very difficult to predict because of the impact of unpredictable events. By employing a methodology that assures

a robust size proxy group, the impact of individual results which appear unreasonable is largely mitigated.

Capital Asset Pricing Model

Q. Please describe the basic theory underlying the CAPM.

A. The principle behind the CAPM theory is that there is no premium, in terms of expected return, for bearing risks that can be eliminated through diversification. According to the CAPM, rational investors will hold a well-diversified portfolio, such that the overall risk of the portfolio, in terms of variability of its returns, closely approximates that of the market as a whole. Thus, the only risk that matters in the CAPM equation is said to be a systematic risk or risk that cannot be diversified away.

Q. What is systematic risk?

A. Systematic risk is a risk common to all equity securities that cannot be eliminated through diversification. It is measured by a variable known as beta. Systematic risk also refers to market risk, which is the uncertainty inherent in the entire market or market segment. Changes in

interest rates, recessions, and wars are all examples of systematic risks.

Q. What is unsystematic risk?

A. Unsystematic risk is the risk that is specific to a particular stock. While it is assumed that most stocks tend to go along with the general market, at least to some extent, factors that are specific to an individual stock are said to affect its unsystematic risk.

Q. How does the CAPM measure an individual stock's risk?

A. According to the CAPM, the appropriate way to measure an individual stock's risk is through a correlation of its return relative to the market as a whole, known as beta. A stock with a beta of 1.0 has a return that mirrors the return of the market, which is usually taken to mean the S&P 500, as a whole. Betas of less than 1.0, which are typical for utility stocks given the moderating influence of regulation and the accompanying perception of reduced risk, indicate that the stocks are less volatile than the market as a whole. Therefore, the CAPM informs us that equity investors will only be compensated for

their actual risk, as measured by beta. Thus, in the case of estimating the return requirements of utility investors, the CAPM result will reflect the degree to which utility stocks are typically less volatile than the market as a whole.

Q. What assumptions is the CAPM based upon?

A. The CAPM model is based on the assumptions that:(1) the capital market is competitive and efficient; (2) investors are risk-averse and demand higher returns for higher risk; (3) market participants can lend and borrow unlimited amounts under the risk-free rate of interest; and (4) investors hold diversified portfolios and operate in capital markets with no transaction costs, taxes, or restrictions on financial transactions. Under the CAPM, the expected return of a security or portfolio is equal to the rate on a risk-free security plus a risk premium multiplied by the asset's systematic risk as indicated by beta.

Q. Please describe the methodology you used to determine your CAPM results.

A. Consistent with the approach that Staff Financial Analysts have employed for many years and which

the Commission has adopted, I averaged the results of two forms of the CAPM: the Traditional CAPM and the Zero-Beta CAPM. The Commission relied on Staff Financial Analysts' ROE methodology that utilized the average results of the two forms of the CAPM in the Order issued on March 25, 2008, in Case 07-E-0523 regarding Consolidated Edison Company of New York, Inc. Also, the Commission applied the same approach in its 2007 Order issued on December 21, 2007, in Case 07-G-0141 involving National Fuel Gas Supply Co.

Q. Why did you use two forms of the CAPM?

A. I use the two forms of the CAPM because the Traditional CAPM model may underestimate the required return when betas are below one. Therefore, to avoid an underestimated return, I use a Zero-Beta methodology as well and average the results of the two approaches. By averaging the results of the Traditional CAPM and the Zero-Beta CAPM, I enhance the accuracy of my overall CAPM ROE determination.

Q. Please describe how a CAPM result is calculated using the Traditional CAPM method.

A. The Traditional CAPM method calculates a required return based on three inputs: the rate of return on a risk-free investment, or Rf, the level of systematic risk for an investment, or β for beta, and the expected market risk premium, or MRP. Typically, the MRP is calculated by subtracting the risk-free rate from the expected market return, or Rm. The form that the Traditional CAPM takes is as follows:

Required Return = Rf + β*(Rm-Rf).

Q. Please describe how a CAPM result is calculated using the Zero-Beta CAPM method.

A. I used the same inputs as the Traditional CAPM methodology; however, instead of multiplying beta by the MRP, as shown in the calculation of the Traditional CAPM methodology, I determined the MRP for the proxy group by multiplying 0.75 by beta by the MRP and adding 0.25 times the MRP. This can be expressed as:

K = Rf + 0.75*β*(Rm-Rf) + 0.25*(Rm-Rf).

Q. How did you calculate the risk-free rate used in your analysis?

A. I estimated the risk-free rate by averaging the 10-year and 30-year Treasury bond yields for the

recent three-month period, May 2023 through July 2023. The result for the three months ended July 2023 is 3.82 percent.

Q. Why do you use both the 10-year and the 30-year Treasury bond yields to estimate the risk-free rate?

A. I use the yields for two different Treasury securities because utility investors generally have both intermediate and long-term investment horizons. To account for these two investment horizons, I use the 10-year and 30-year Treasury securities to calculate my result. In the past, the Commission has adopted this approach. For instance, in the June 17, 2011, Order Establishing Rates for Electric Service, in Case 10-E-0362 regarding Orange and Rockland Utilities, Inc., the Commission stated its preference for this approach. Specifically, on page 75 of the June 17, 2011, Order, the Commission stated that "using a combination of treasury yields is consistent with our practice and supported by the varying nature of investor holding periods."

Q. Why are you using a three-month average of the

Treasury yields for the risk-free rate calculation?

A. I consider the use of a three-month average to be appropriate because it smooths out potential short-term volatility. Moreover, the Commission employed three-month average yields in its April 24, 2009, Order Setting Electric Rates in Case 08-E-0539, concerning Consolidated Edison Company of New York, Inc. for consistency with the three-month timeframe employed in its DCF cost of equity determination. To maintain this consistency, since I am employing the most recent three months of market data in my DCF calculation, I used the three-month average Treasury yield data in the CAPM analysis as well.

Q. How did you determine the beta for your CAPM analyses?

A. I calculated the mean, or average, beta reported by Value Line for each of the 31 companies in my proxy group. The calculated mean beta of my proxy group is 0.889, rounded up to 0.89.

Q. How did you determine the appropriate cost of market return to use in your CAPM analyses, and what was the result?

A. Because the cost of equity is, by its nature, a forward-looking concept, I employed an ex-ante or forecasted analysis, relying upon the required market return estimates for the S&P 500, published monthly by Bank of America Securities Quantitative Profiles, or BAQP, formally known as Merrill Lynch Quantitative Profile, in its Quantitative Profile reports. My market rate of return is derived from the two forward-looking returns on the market: the required return and the implied return. Specifically, I used the monthly editions of the Quantitative Profile reports for June 2023, July 2023, and August 2023 editions, and averaged the required and implied market returns of each of the three point-in-time forecasts, to arrive at a market return of 10.85 percent. Excerpts of these publications are included in Exhibit___(KXD-24).

Q. Why are you using the most recent three months of BAQP's expected market returns in your calculations?

A. I use expected market return estimates from the most recent three months to be consistent with the timeframes of the other data inputs used in

my CAPM and DCF calculations. By matching the timeframe upon which my other parameters are calculated, I arrive at the most appropriate estimate of the required market rate of return to use.

Q. How did you derive your MRP?

A. The MRP is the difference between the expected market return on the equity market and the return required on a risk-free investment. Given my risk-free rate of 3.82 percent, I calculated the expected MRP to be 7.03 percent by subtracting the risk-free rate from the 10.85 percent expected market return.

Q. Why didn't you utilize an ex-post method to derive the MRP?

A. The application of an ex-post, or historical, MRP method is problematic because ex-post MRPs are based on the premise that past performance is a valid proxy for expectations regarding future results. Yet, the underlying theory of this approach has never been substantiated. The MRP is essentially a mean-reverting measure. Furthermore, this approach has been shown to periodically produce counterintuitive results.

Q. Do you have any other concerns about the ex-post, or historical MRP?

A. Yes. This approach has also been known to be highly sensitive to the actual period selected to calculate the premium. For example, from time to time, the actual market return can become negative due to many factors, including recessions, war, and/or depressed earnings. The COVID-19 pandemic is an example of this kind of event risk. Therefore, while investors expect the market return to be positive over the long run, they also expect the market return to fluctuate in the short term, punctuated by periods of negative return. Due to these structural shifts in the economy, the application of an expected long-term market return must consider realistic growth rates and current economic factors.

Q. Why did you use BAQP data to calculate the MRP in your CAPM analyses?

A. The Commission has consistently applied and implemented this market risk premium methodology since its October 3, 1996, Opinion and Order Concerning Revenue Requirement and Rate Design in

Case 95-G-1034, concerning Central Hudson. In that Order, the Commission approved the use of the Merrill Lynch estimate stating, "…the Judge's market return calculation based on Merrill Lynch data is a reasonable method of deriving a risk premium."

Q. Please explain how the calculation of the MRP can be problematic.

A. The alternative to an MRP that is historically derived, is a forward-looking one. While I advocate in favor of using an expected MRP in my CAPM methodology, I also acknowledge that such an approach is, by necessity, subject to a substantial amount of judgment. This is one of the principal reasons that Staff Financial Analysts have consistently argued that the CAPM should only be accorded half the weight of the multi-stage DCF-derived cost of equity estimate.

Q. What is the Traditional CAPM result using your stated inputs?

A. I determined the Traditional CAPM result to be 10.08 percent, calculated as follows:
3.82% + [0.89 * (10.85% - 3.82%)] = 10.08%.

Q. What is the Zero-Beta CAPM result using your

stated inputs?

A. The Zero-Beta CAPM is 10.27 percent, calculated as follows:
3.82% + [0.75 * 0.89 *(10.85% - 3.82%)] + [0.25 * (10.85% - 3.82%)] = 10.27%.

Q. Please explain how you used the results of these two CAPM methods in your calculation of the ROE for your proxy group.

A. I averaged the results of the Traditional and Zero-Beta CAPM calculations to arrive at a determination of 10.18 percent. This is the same approach that has been adopted or approved by the Commission in rate cases for many years. The 10.18 percent CAPM is illustrated in Exhibit___(KXD-25).

Return On Equity Conclusion

Q. How did you determine your recommended 9.1 percent cost of equity for the Companies?

A. I determined my recommended 9.1 percent cost of equity by weighting: (1) the 8.52 percent average DCF result from my proxy group two-thirds; and (2) the 10.18 percent average of the two CAPM results one-third. The resulting ROE is 9.07 percent, rounded up to 9.10, as illustrated in

Exhibit___(KXD-20).

Q. Do you recommend that your ROE be updated during the course of these rate proceedings?

A. Yes. The Commission should update the ROE estimate before issuing its decision to reflect the most current financial data available.

Discussion of Companies' Witness Nowak's ROE Approach

Q. Please summarize the approach followed by Companies' witness Nowak.

A. Companies' witness Nowak began his analysis by applying his cost of equity financial models to a proxy group of 19 utility holding companies from a universe of 47 publicly traded holding companies that Value Line categorized as "electric utilities" and "natural gas utilities." Next, Companies' witness Nowak developed three multi-stage DCF models utilizing the low, mean, and high earnings growth rate estimates published by Yahoo Finance, Zack's, and Value Line as shown in his Exhibits___(JCN-1) and (JCN-4). These DCF calculations resulted in a low, mean, and high ROE estimate for each company in his proxy group.

Q. What CAPM did Companies' witness Nowak utilize to determine the Companies ROE?

A. Companies' witness Nowak utilized both the Traditional form of the CAPM and the Zero-beta CAPM applied to both Bloomberg and Value Line betas which resulted in two sets of three ROEs for his proxy group.

Q. What ROE did Companies' witness Nowak recommend?

A. Companies' witness Nowak concluded that an appropriate ROE for the Companies ranges between 9.8 percent and 10.7 percent.

Q. What ROE have the Companies requested in these proceedings?

A. The Companies are seeking a 9.8 percent ROE for the Rate Year.

Q. Do you have any concerns with Companies' witness Nowak's analysis?

A. I am concerned with three primary aspects of Companies' witness Nowak's analysis: (1) the composition of his proxy group; (2) the use of excessive growth rates in his DCF analyses; and (3) the use of several flawed key inputs employed in his CAPM analyses, particularly his market return estimate, and to a lesser extent the risk-free rate. Finally, I strongly disagree with Companies' witness Nowak's premise that current

market conditions warrant that the CAPM be accorded equal weighting with the DCF.

Criticism of The Proxy Groups Employed by Companies' Witness Nowak

Q. How did Companies' witness Nowak develop his proxy group of 19 utility holding companies?

A. Beginning with the 47 companies classified as "electric utilities" and "natural gas utilities" by Value Line, he applied the following screening criteria to eliminate companies that: (1) are not covered by at least two utility industry equity analysts; (2) do not have investment grade corporate credit ratings and/or senior unsecured bond ratings according to S&P and Moody's; (3) have not paid regular dividends or do not have positive earnings growth projections from at least one source; (4) derive less than 70 percent of total operating income from regulated utility operations; (5) are known to be party to a merger, acquisition, or other transformational transaction; and (6) own generation comprising greater than 60 percent of the company's megawatt-hours, or MWh.

Q. What is the impact of Companies' witness Nowak's

selection criteria on his proxy group composition?

A. In response to DPS-711, Companies' witness Nowak indicated that, based on his six selection criteria, he eliminated 28 companies from the Value Line universe of 47 electric and natural gas companies in constructing his proxy group.

Q. Do you have concerns with any of Companies' witness Nowak's selection criteria?

A. Yes, I have two primary concerns regarding his selection criteria relating to net operating income and companies owning generating assets responsible for more than 60 percent of MWh sales.

Q. Please explain the reasoning behind these concerns.

A. I find Companies' witness Nowak's requirement that a company must have 70 percent of its net operating income from regulated utility operations to be too restrictive. Operating income is the amount of revenue left over after accounting for all the expenses necessary to keep the business running.

Q. What is the impact of this methodology on the

resulting proxy group?

A. His methodology results in the elimination of companies whose risk characteristics are closely comparable to KEDNY and KEDLI. For example, Sempra Energy, which is in my proxy group was eliminated from his proxy group using the 70 percent operating income criteria. Moreover, using operating earnings instead of a revenue criterion for establishing the proxy group may allow companies into the proxy group that contain substantially greater risk than a typical regulated utility. Revenues more accurately reflect the exposure to riskier competitive operations, because it is not uncommon for competitive ventures to be unprofitable at times. Accordingly, it is quite possible that using the 70 percent of operating income criteria could lead to the selection of companies that have a higher inherent risk profile than is proper for a regulated utility proxy group.

Q. Is there any other reason why it is inappropriate to use operating income as a selection criterion?

A. As illustrated in Exhibit___(KXD-26), the difference in accounting policy choices, such as

a difference in inventory valuation policy options, and a difference in depreciation choices across companies, could have a more significant impact on reported operating income than on reported revenues. For these reasons, revenue remains a superior criterion for ensuring that a utility proxy group properly reflects the amount of regulated business that most closely approximates the risk of the regulated utility being analyzed.

Q. What is your recommendation concerning Companies' witness Nowak's use of operating income?

A. I recommend that Companies' witness Nowak's use of operating income as a criterion be rejected, and instead, my 70 percent regulated revenue criterion be adopted.

Q. Please explain your second concern regarding Companies' witness Nowak's selection criteria.

A. Companies' witness Nowak's selection criteria is unreasonably restrictive in limiting his proxy group to primarily transmission and distribution utilities by excluding those having owned generation compromising greater than 60 percent of MWh sales to ultimate customers. This

restriction resulted in the elimination of eleven companies, that are all comparable in risk to KEDNY and KEDLI, and would be suitable surrogates for the Companies.

Q. What is the result of Companies' witness Nowak's selection criteria?

A. By drastically reducing the size of his proxy group, Companies' witness Nowak has undermined the reliability of his results. The eleven eliminated companies, which are all in my proxy group are: Alliant Energy Corporation, Ameren Corporation, CenterPoint Energy, Inc., Duke Energy Corporation, Entergy Corporation, IDACORP, Inc., NextEra Energy, Inc., Pinnacle West Capital Corporation, Portland General Electric Company, Southern Company, and Wisconsin Energy Corporation.

Criticism of Companies' witness Nowak's DCF Methodology

Q. Please describe Companies' witness Nowak's DCF methodology.

A. Companies' witness Nowak uses a three-stage DCF analysis based on a three-month average stock price and a range of near-term growth rate

assumptions that produces a mean ROE of 9.68 percent. Specifically, his analysis relies on equity analysts' earnings growth forecasts published by Value Line, Zacks, and Yahoo Finance.

Q. Do you have any concerns with Companies' witness Nowak's DCF analysis?

A. I have several concerns. Foremost among them, I find his third-stage growth rate of 5.46 percent to be excessive. Also, I find his first-stage growth rates to be inferior to mine. Companies' witness Nowak's use of an excessive third-stage growth rate of 5.46 percent is the principal reason that his 9.68 percent mean DCF-based ROE result is flawed. While the use of a three-stage DCF growth model is not necessarily problematic, his use of an excessive growth rate in his long run or third-stage growth rate, and to a lesser extent, some of the excessive growth rates he uses in the first stage of his analysis are problematic.

Q. Please explain why you believe that his first-stage growth rates are problematic.

A. Companies' witness Nowak's DCF model uses

earnings growth rate estimates to determine short-term dividend growth rates, ignoring available dividend growth rates or forecasts. This is in direct contrast to the basic premise of the DCF which measures the present value of future dividends. In using earnings growth, Companies' witness Nowak has simply assumed that dividend growth will match earnings growth but has presented no evidence why this expectation is valid. Of greater concern is his use of a 17.21 percent EPS growth rate for PPL Corporation. I believe that a growth estimate of that magnitude is unreasonable and that he should have ignored it in his analysis.

Q. What would have been the impact of ignoring the 17.21 percent growth rate for PPL Corporation on the DCF results?

A. The impact of ignoring this 17.21 percent EPS growth rate would have reduced Companies' witness Nowak's DCF results by nine basis points from 9.68 percent to 9.59 percent.

Q. Do you have any other concerns about Companies' witness Nowak's growth rates?

A. Yes, I also have a concern about his use of a

historically derived real GDP rate, together with a forecasted inflation rate as the long-term growth rate of his proxy group.

Q. How did Companies' witness Nowak calculate the GDP growth rate?

A. Instead of using estimates of a future GDP growth rate, Companies' witness Nowak utilized a 3.16 percent historical average growth rate in real GDP for the period 1929 through 2022 based on data from the Bureau of Economic Analysis. Companies' witness Nowak then calculated a 5.46 percent forecasted nominal GDP rate by taking this historical average growth rate, together with his expected average inflation rate of 2.23 percent. The 2.23 percent expected average inflation rate was calculated based upon an average of 2.10 percent expected inflation published by the Blue Chip Financial Forecasts, 2.35 percent Consumer Price Index, or CPI, and 2.25 percent GDP deflator, both of which were derived using data from the EIA. The derivation of his 5.46 percent long-term GDP growth rate is shown on page one of his Exhibit___(JCN-2).

Q. Do you agree with Companies' witness Nowak's 5.46

percent long-term growth rate estimate?

A. I do not think the use of his historically derived real GDP as the long-term growth rate is appropriate. Historical growth rates are necessarily measured over a subjective period, and in this instance, one that incorporates far different circumstances than exist today. Historical average growth rates by their nature vary completely, and inappropriately ignore economic conditions today. In addition, his use of a growth rate based on the average GDP over the past 93 years, from 1929 to 2022, does not reflect a realistic estimate of a sustainable growth rate for a company in the utility industry. The utility sector is a mature and stable industry; thus, it typically would be expected to grow at a slower rate than the overall GDP.

Q. Are there estimates of the long-run growth in the economy that you believe provide a more reasonable estimate of long-run growth?

A. Yes, I am aware of two reasonable estimates of the long-run growth rate. The first source is the Long-Range Consensus U.S. Economic

Projections provided by Blue Chip Economic Indicators, which I previously discussed, and employed in ascertaining the reasonableness of my sustainable growth rate. Not only does this report project twice as far out into the future as any other reputable source of economic data, but it also reflects the forward-looking consensus of approximately 50 of the financial community's most prominent economists.

Q. What does Blue Chip Economic Indicators most recent publication state regarding long-term GDP growth?

A. According to the March 10, 2023, publication, which is the most recent, the consensus long-run nominal GDP growth rate for the most distant period forecast, 2030-2034 is 3.9 percent, which includes both 1.8 percent real GDP growth and 2.1 percent expected inflation components.

Q. What do you think of the Blue Chip Economic Indicator's forecast?

A. A reasonable investor would expect a similar long-term growth rate for an energy utility rather than the excessive 5.46 percent growth rate used by Companies' witness Nowak.

Q. What is the second estimate of the long-run growth in the economy that you believe is a more reasonable estimate of long-run growth?

A. The second source as shown in my previously mentioned Exhibit___(KXD-23), is based on information and data from EIA. On page four of its March 2023 Annual Energy Outlook, three possible paths for U.S. economic growth are mentioned. These are the "Reference," "High Economic Growth," and "Low Economic Growth" scenarios. In the High Economic Growth case, the average annual growth in real GDP is 2.3 percent from 2022 to 2050, compared with 1.9 percent in the Reference case. The Low Economic Growth case assumes a lower rate of annual growth in real GDP of 1.4 percent. The EIA's optimistic long-run real GDP growth rate of 2.3 percent suggests a high nominal GDP growth rate of about 4.5 percent (2.4 percent plus 2.2 percent inflation).

Q. What does this forecast indicate about Companies' witness Nowak's projected long-term growth rate?

A. In light of this information, Companies' witness Nowak's use of an excessively high long-term growth rate of 5.46 percent undermines the

reliability of his DCF results.

Q. Please describe how Companies' witness Nowak derives his short-term dividend growth rates.

A. In the first stage of the three-stage DCF model, Companies' witness Nowak takes the current annualized dividend for each company in his proxy group and escalates the dividend for a period of five years based on the average of the three-to-five-year earnings growth estimates reported by Yahoo Finance, Zacks, and Value Line.

Q. Why is this approach problematic?

A. It is highly unlikely that investors would rely exclusively on the earnings per share growth rate forecasts of Wall Street analysts in determining short-term dividend projections. In support of his use of earnings growth rate, Companies' witness Nowak, on pages 36 to 38 of his testimony, cited a 25-year-old article written by Dr. Robert Harris, stating that a "growing body of knowledge shows that analyst earnings forecasts are indeed reflected in stock prices." While I agree that all relevant information including earnings forecast is incorporated into a company's stock price, the linear relationship

of earnings to dividend growth that he assumes is unlikely. That is, payout ratios will change based on many factors including individual company cash flow requirements, future market conditions as well as other factors. Investors will not just look at expected earnings but also factor in all relevant information when estimating their required return.

Q. Does Companies' witness Nowak have any concern with the use of Value Line short-term dividend projections?

A. Yes, he has some concerns. Companies' witness Nowak, on page 39 of his testimony, claims that since Value Line's projections of short-term dividend growth do not explicitly include growth in retained earnings, their use is improper as inputs to the DCF analysis. What he has failed to recognize is that, while in the long-term, dividend growth is driven by the combination of retention growth and issuing stock above/below book value; in the short run, dividend policy can change and result in dividends growing above or below retention growth. Consequently, his short-term dividend projections are a direct product of

the average earnings growth estimates of three different publications, without any consideration being given to the growth rates' effect on future dividend payouts. My use of Value Line dividend growth projections recognizes the impact of payout policy while his sole use of short-term earnings forecasts does not.

Q. Does Companies' witness Nowak have other criticisms of using only Value Line dividend growth rates?

A. Yes. Companies' witness Nowak claims Value Line's dividend projections are the growth expectations of a single analyst. Further, he attempts to discredit the utilization of Value Line by claiming the publication is based on a single analyst's viewpoint.

Q. What is your response to Companies' witness Nowak's concerns regarding Value Line?

A. While Value Line assigns a lead analyst in the electric utility industry who is then credited for each company's report, each report that Value Line develops goes through a continual evaluation and quality control process where multiple analysts review each report before they are ever

posted. The Value Line document describing this process is included in Exhibit___(KXD-27).

Q. Is there any advantage to the continued use of Value Line projections in your CAPM and DCF analysis?

A. Yes. Staff Financial Analysts and the Commission have consistently relied on Value Line for many years. This continued use of Value Line is generally understood by the investment community and lends a degree of predictability to the rate-setting process in New York State. While the sole reliance on Value Line has been criticized in the past by some companies' ROE witnesses, it is important to note that its continual use partly contributes to the predictability in the level of allowed equity returns that equity investors in New York utilities can expect. This is particularly important to the major credit rating agencies, who view unpredictability as an additional risk.

Q. What is your view on the potential impact of higher inflation and Federal Reserve monetary policy on the Companies' cost of equity and their ability to have access to the capital markets at

reasonable terms?

A. Stock prices reflect the collective judgment of all stock market participants, encompassing known factors that influence stock prices, such as the Federal Reserve's monetary policies and expected inflation. As discussed earlier, my DCF and CAPM cost of equity approaches consider prevailing and expected market conditions. Therefore, I believe that the Companies' cost of equity and their ability to access capital at reasonable terms should not be hindered by these market conditions. Moreover, my cost of capital recommendations will allow the Companies to maintain their credit ratings and secure capital at reasonable terms.

Q. Are you aware of any financial literature that supports the notion that stock prices reflect the collective judgment of all stock market participants, including the Federal Reserve's monetary policies and expected inflation?

A. Yes. In his book titled, "The Cost of Equity - A Practitioner's Guide (2010 Edition)," David C. Parcell discusses the significance and interpretation of stock prices. On pages 88 and

89, Parcell states that "the market price of a firm's stock represents the collective judgment of all stock market participants regarding the value of a firm at a given point in time. It encompasses their interpretation of various relevant factors, such as past, present, and future earnings, earnings risk, dividend policy, and other pertinent aspects." This excerpt from pages 88 and 89 of the book is included as Exhibit___(KXD-28).

Q. What is your overall conclusion of witness Nowak's DCF approach?

A. My overall conclusion is that Companies' witness Nowak's use of a historically derived long-term growth rate, 17.21 percent earnings per share growth rate for PPL Corporation, and his analysis of the market conditions impact on the Companies' cost of equity, are not reasonable and should therefore, be rejected by the Commission.

Criticism of Companies' witness Nowak's CAPM Methodology

Q. Would you please summarize Companies' witness Nowak's CAPM approaches?

A. Companies' witness Nowak provided a total of 12

CAPM-derived ROE estimates for his proxy group using the same CAPM methodologies that I employ. Companies' witness Nowak calculated six ROEs using the Traditional CAPM methodology. Also, witness Nowak calculated six ROEs using the Zero-beta CAPM methodology, with two different beta sources in combination with three different risk-free rate estimates for each version of the CAPM.

Q. Please explain how Companies' witness Nowak derived each of the three major inputs necessary for his CAPM methodologies: the risk-free rate, beta, and the market risk premium.

A. As shown in his Exhibit___(JCN-4), Companies' witness Nowak's Traditional and Zero-beta CAPM methodologies use a risk-free rate based on the current three-month average yield from January 2023 to March 2023 of 30-year Treasury bonds averaging 3.75 percent, the near-term projected 30-year Treasury yield for the five forward-looking quarters, which encompasses the second quarter of 2023 through the second quarter of 2024, averaging 3.88 percent, and the long-term projected 30-year Treasury yield for the period 2024-2028, averaging 3.90 percent.

Q. How did Companies' witness Nowak develop the market risk premium?

A. Companies' witness Nowak developed a market risk premium based on an estimated market return of 12.27 percent and the subtraction of each of the three risk-free rates as discussed earlier. Companies' witness Nowak utilized two different beta determinations within each of his CAPM methodologies. Companies' witness Nowak applied a 0.812 Bloomberg average beta and 0.881 Value Line average beta.

Q. What were the results of Companies' witness Nowak's analysis?

A. Companies' witness Nowak developed three Traditional CAPM estimates of the cost of common equity ranging from 10.96 percent to 10.99 percent, and three zero-beta ROE estimates ranging from 11.29 percent to 11.31 percent. By averaging all these results, Companies' witness Nowak's CAPM methodology produced low, mean, and high CAPM-based ROE estimates of 11.13 percent,11.14 percent, and 11.15 percent, respectively.

Q. Do you have concerns with Companies' witness

Nowak's CAPM analyses?

A. Yes, I have three principal concerns regarding Companies' witness Nowak's CAPM analyses. Listed in order of magnitude of their impact, these concerns relate to: (1) his derivation of a 12.27 percent market return; (2) his beta determinations; and (3) his risk-free rate. Collectively, the impact of these inputs results in an overstatement of his CAPM-based cost of equity estimate of about 140 basis points.

Q. How did Companies' witness Nowak derive the 12.27 percent market return?

A. Companies' witness Nowak first derived a 14.08 percent market return by applying a single-stage DCF analysis to the companies in the S&P 500. He then calculated a 10.47 percent Merrill Lynch market return and averaged the two results, resulting in a 12.27 percent market return. For his market return from the Bank of America Quantitative Profiles, he utilized the monthly editions of the publication from December 2022 to February 2023, averaging the required and implied market returns of each of the three point-in-time forecasts, resulting in a market return of 10.47

percent.

Q. Please explain your concern with the approach he used to determine his 12.27 percent required market return.

A. While I have no concern with the Bank of America Quantitative Profiles market return approach used by Companies' witness Nowak, I do have a concern regarding the use of a constant growth DCF analysis of the S&P 500 to determine a 14.08 percent market return. His use of the excessively high 14.08 percent market return versus my 10.80 percent market return is the main reason for the large differences between his CAPM results and mine. His excessive 14.08 percent growth rate relies on Bloomberg and Yahoo Finance earnings growth rate estimates, which is very problematic.

Q. Explain how Companies' witness Nowak's reliance on a 14.08 percent growth rate is problematic.

A. Quite simply, the basic assumption of this approach is that the Bloomberg and Yahoo Finance earnings growth rate estimates formulated for the next three-to-five years will last into perpetuity, which is not reasonable. This is

precisely why Staff Financial Analysts rely on the ex-ante estimate of the required return of the S&P 500 provided by Bank of America Quantitative Profiles, as shown in Exhibit___(KXD-24). Bank of America Quantitative Profiles' multi-stage DCF-derived required return does not make this unrealistic assumption but rather uses three specific stages with varying durations depending on the facts and circumstances of the company being examined.

Q. Why is Companies' witness Nowak's approach unreasonable?

A. The unreasonableness of using a constant growth DCF calculation to estimate the required market return is perhaps best illustrated by considering the fact that 38 of the S&P 500 companies in Companies' witness Nowak's market return calculation have near-term earnings growth estimates in excess of 20.0 percent and range as high as 81.14 percent as shown in his Exhibit___(JCN-5). It is plainly unreasonable that investors would assume that those companies would be able to maintain those extraordinary growth rates forever.

Q. Does Companies' witness Nowak's market return calculation have any other flaws?
A. Yes. Many of the companies in the calculation of his excessive 14.08 percent market return are not paying any dividends. The DCF calculation is premised upon the present value of future cash flows, in this case, dividends. If companies are not paying dividends, then the whole DCF calculation is flawed because a DCF cannot be performed. Therefore, his 12.27 percent market return should be rejected because of flaws in his calculation.
Q. What is the impact of using solely Companies' witness Nowak's 10.47 percent Bank of America Quantitative Profiles market return rather than combining it with his excessive 14.08 percent market return?
A. The impact is an overestimation of about 156 basis points in his CAPM-based ROE results.
Q. Please explain your second concern regarding the derivation of Companies' witness Nowak's beta estimates.
A. The Commission has consistently utilized Value Line betas, and one of the principal reasons for

doing so is because of Value Line's superiority over other sources like Bloomberg.

Q. Are there any independent sources indicating that Value Line betas are superior?

A. Yes. In their book entitled, "Capital Structure and Corporate Financing Decisions: Theory, Evidence, and Practice," H. Kent Baker and Gerald S. Martin wrote about the superiority of Value Line betas over Bloomberg betas. They wrote that "Value Line's adjustments also appear informed by more than the type of simple smoothing done by Bloomberg." After plotting historic betas, Value Line adjusted betas, and Bloomberg adjusted betas, the authors concluded that analysts "be wary of blindly using historic betas from individual company data." Further, the authors stated, "[t]he tails of the distribution for historic betas contain significant measurement errors. A clear advantage of Value Line betas (or those from other providers who do more than just historic regression) is that they reflect professional judgment and attention to detail." This report is attached as Exhibit___(KXD-29).

Q. Has the accuracy of Value Line betas been tested?

A. Yes, it has been tested. In her empirical research work entitled, "Whose Beta is Best?" Ms. Diana Harrington investigated the accuracy of Value Line betas and those betas reported by Merrill Lynch, Wilshire Associates, Rosenberg and Associates (now BARRA), and others.

Q. What did Ms. Harrington's analysis determine?

A. For the period studied, Value Line consistently outperformed all other services over the forecast horizon of one to four years for public utilities. The study also indicated that increased data points increase the beta accuracy. The report is attached as Exhibit___(KXD-30). Comparatively, I have not come across any empirical research work that shows that Bloomberg beta estimates are superior to Value Line betas.

Q. Please explain your third concern relating to Companies' witness Nowak's sole use of the 30-year Treasury yield as the appropriate risk-free rate.

A. Companies' witness Nowak argues that the sole use of the yield on the 30-year Treasury is appropriate because, in his view, utility companies represent long-duration investments.

Yet, it has long been Commission policy to rely on the average of the 10-year and 30-year Treasury yields to arrive at the risk-free rate as investors have a range of investment horizons. Furthermore, as the Commission has maintained for many years, current rates are the best indicator of future rates as they are based upon the latest available information to investors. The rationale for this approach is well-established and reflects the reality that there are investors with intermediate term as well as long-term investment horizons.

Q. Did Companies' witness Nowak cite a Morningstar article to suggest that the Commission's use of a blend of treasury yields is flawed?

A. Companies' witness Nowak cited the Morningstar article to support his argument that the Commission's preferred approach of using a blend of 10-year and 30-year Treasury yields is flawed because it does not address the expected lives of the Companies' assets or the equity duration of the utility industry. In the article, Morningstar states that "the time horizon of the chosen Treasury security is that it should match

the time horizon of whatever is being valued." With respect to the equity duration and its relationship with the Treasury yield time horizon, Companies' witness Nowak specifically mentioned that "because utility companies represent long-duration investments, it is appropriate to use yields on long-term Treasury bonds as the risk-free rate component of the CAPM."

Q. Do you agree with Companies' witness Nowak's argument?

A. No, I do not. While Companies' witness Nowak is correct that utility plant assets have very long lives, his conclusion that all utility equity investors have an investment horizon of 30 years is unsubstantiated and erroneous. One needs to look no further than the long-term debt obligations supporting the Companies own rate bases to understand that investors have different time horizons.

Q. Is Companies' witness Nowak's use of Treasury forecasted yields appropriate for use in the CAPM?

A. No, it is not. His use of Treasury forecasted

yields should be rejected. Staff Financial Analysts and the Commission have maintained for many years that current interest rates are the best indicator of future interest rates as such rates are based on the latest available information to investors.

Q. What is the combined effect of a blended yield of 10-year and 30-year Treasury yields on Companies' witness Nowak's CAPM results?

A. This is similar to the Commission's preferred approach. The impact is an overestimation of about 13 basis points in his CAPM analysis.

Q. What is your overall conclusion relating to your concerns about Companies' witness Nowak's excessive 12.27 percent market return, his beta determinations, and risk-free rate derivation?

A. My overall conclusion is that if he had ignored his excessive 12.27 percent market return, disregarded his Bloomberg average betas, his 3.88 percent, and 3.90 percent forecasted risk-free rates, his mean CAPM result would have been completely different. If Companies' witness Nowak had instead relied on his Bank of America Quantitative Profiles average market return of

10.47 percent, his 0.881 Value Line average beta, and his 3.75 percent then current risk-free rate, his mean CAPM-based cost of equity estimate would have been 9.78 percent compared to his 11.14 percent mean CAPM result shown in a chart on page five of his testimony. This indicates an overstatement of his CAPM-based ROE estimate by approximately 140 basis points. Therefore, I recommend that the Commission reject Companies' witness Nowak's 12.27 percent market return, his Bloomberg betas, and forecasted risk-free rate derivation.

Equal Weighting of DCF and CAPM Results

Q. Why does Companies' witness Nowak argue that the CAPM methodology should be accorded an equal weighting with the DCF methodology?

A. Companies' witness Nowak's explanation to justify his equal weighting is based on two main factors: (1) the effect of the Federal Reserve monetary policies on market conditions; and (2) the reliance of the CAPM on observable market data such as risk-free rate, betas, and the market risk premium. On page 55 of his testimony, Companies' witness Nowak concluded that "since

the CAPM directly accounts for changes in interest rates, it is appropriate to give the CAPM more weight as compared to recent Commission decisions in the current capital market environment… [and] it is reasonable to apply equal weighting to the DCF and CAPM methods when determining the ROE for the Companies."

Q. What is your response to Companies' witness Nowak's argument?

A. My response is that the CAPM should be accorded a lesser weight than the DCF.

Q. Why should the CAPM be weighted less than the DCF?

A. The CAPM presents a conceptual framework that provides a reasonable estimate of a firm's cost of equity. Yet, given some of the perpetual weaknesses of the CAPM, as discussed previously in my testimony, the Commission should continue its CAPM weighting of one-third compared to the DCF weighting of two-thirds.

Q. Has the Commission ever addressed the issue of equal weighting of the DCF and CAPM?

A. Yes. The Commission has consistently used the DCF as its primary methodology for determining

ROE by applying a two-thirds weighting in the determination of the cost of equity. About a decade and a half ago, some company witnesses including Dr. Morin's ROE testimony filed on May 9, 2008 in Case 08-E-0539 involving Consolidated Edison Company of New York, Inc., raised concerns about the weighting accorded the DCF and argued for equal weighting of the DCF and CAPM when the market-to-book ratio exceeds one or when the DCF was producing lower results than other methodologies the witnesses were advocating. The Commission ultimately decided in its April 24, 2009, Order in Case 08-E-0539, on page 133 to support a cost of equity determination with a two-thirds DCF weight and one-third CAPM weight.

Q. Would you reference some of the cases where the Commission stated its preference for the two-thirds and one-third weighting of the DCF and CAPM?

A. The Commission affirmed the two-thirds DCF and one-third CAPM weighting in its March 25, 2008, Order in Case 07-E-0523 regarding Consolidated Edison Company of New York, Inc. Also, the Commission affirmed this weighting in its 2007

Order issued on December 21, 2007, in Case 07-G-0141 involving National Fuel Gas Supply Co. In the 2007 Order, on page 40 the Commission stated, "[w]e also agree with Staff, CPB, and Multiple Interveners that the Company (NFG) has not provided any compelling reasons to provide equal weight to the DCF and the CAPM methods... Accordingly, we will continue to use the two-thirds DCF Method and one-third CAPM method weighting in this case."

Q. Was there any other case where the Commission supported a similar weighting to the DCF and CAPM results that should be referenced?

A. Yes, on pages 14 and 15 of its Order issued on October 18, 2007, in Cases 06-E-1433 and 06-E-1547, Orange & Rockland Utilities, Inc., the Commission stated, "[w]e will continue to accord two-thirds weight to the DCF result and one-third to the CAPM result as we have in past decisions." Also, it is worth noting that the Commission affirmed the proportionate weighting of the DCF and the CAPM when the utility stock valuations and dividend yields were similar to current levels.

Q. What is your conclusion based on these Commission rulings?

A. I believe that Companies' witness Nowak has not presented any new evidence that would compel the Commission to reject its long-standing policy of a two-thirds and one-third weighting of the DCF and CAPM, and as such Companies' witness Nowak's approach should be rejected.

Impact of Correcting the Flaws in Companies' witness Nowak's ROE Approach

Q. What would be the impact on Companies' witness Nowak's results if he addressed the principal flaws in his DCF and CAPM approach?

A. The combined effect of correcting the flaws in his DCF and CAPM approach including the elimination of the excessive growth rate of PPL Corporation, relying solely on the Value Line betas, and weighting his DCF and CAPM results at two-thirds and one-third, respectively, is a reduction of 85 basis points from the 9.8 percent ROE to 8.95 percent. This is close to my recommended 9.1 percent ROE. This calculation is shown in Exhibit___(KXD-31).

Regulatory Risk Assessment in New York

Q. Do you agree with Companies' witness Nowak's conclusion concerning regulatory risk assessment?

A. No, I do not agree with Companies' witness Nowak's conclusion. Companies' witness Nowak concluded that the Commission's efforts in implementing risk-reducing measures have resulted in no significant difference in business risk between New York and other jurisdictions within his proxy group. While some jurisdictions have implemented similar risk-reducing policies, few have matched the level of New York State's risk-reducing measures.

Q. Did Companies' witness Nowak provide a list of risk-reducing measures implemented by other regulatory bodies for the operating subsidiaries contained in his proxy group?

A. Yes. In his Exhibit___(JCN-6), Companies' witness Nowak listed four risk-reducing mechanisms consisting of a fully forecasted test year, fuel/purchased power cost recovery mechanism, revenue decoupling mechanism, and capital cost recovery mechanism.

Q. What does your analysis of Companies' witness

Nowak's proxy group indicate about his assessment of these risk-reducing mechanisms in relation to New York's risk-reducing policies?

A. My analysis of Companies' witness Nowak's Exhibit___(JCN-6) shows that out of the 91 operating utilities held by his proxy group, only two, Edison International in California and Hawaiian Electric Industries, Inc. in Hawaii, have implemented fully forecasted test years, fuel/purchased power cost recovery mechanisms and revenue decoupling mechanisms. Interestingly, only one operating subsidiary, Hawaiian Electric Industries, Inc, has all the risk-reducing mechanisms, consisting of a fully forecasted test year, fuel/purchased power cost recovery mechanism, revenue decoupling mechanism, and capital cost recovery mechanism, shown in his Exhibit___(JCN-6). Meanwhile, his own exhibit shows that both KEDNY and KEDLI have implemented all of these risk-reducing mechanisms.

Q. Are you aware of any recent commentaries from the investment community or industry reports that evaluated the utility regulatory environment in the United States?

A. Yes, I am aware of a Regulatory Research Associates, referred to as RRA, recent publication about the utility regulatory environment in the United States. RRA consistently evaluates the regulatory climates of utility companies in the United States. RRA takes into account factors such as authorized and earned return on equity, capital structure, and alternative rate-making policies when assessing regulatory environments. This report is shown as Exhibit___(KXD-32).

A. How does RRA categorize and rank regulatory jurisdictions?

Q. RRA employs three primary rating categories: "Above Average," "Average," and "Below Average." Within each category, rankings from "1" to "3" indicate relative positions, with "1" representing the strongest. RRA employs a total of nine rating categories, spanning from the most favorable climate of "Above Average/1" to the least favorable climate of "Below Average/3."

Q. What ranking has RRA assigned to New York?

A. RRA has assigned New York a ranking of "Average/2." This designation is consistent with

the majority of the regulatory jurisdictions across the United States, as shown on page eight of the report in Exhibit___(KXD-32).

Q. Is there any other RRA publication or profile that discusses New York's regulatory environment in detail?

A. Yes, in March 2023, RRA published a profile about the New York State Public Service Commission. In the profile, shown as Exhibit___(KXD-33), RRA notes that while the Commission generally authorizes electric and gas return on equity values lower than industry averages, these decisions often stem from multi-year settlements that incorporate increasing rate bases, revenue decoupling mechanisms, and deferral accounting. Additionally, the use of fully forecasted test periods and other factors in the rate-setting process enhance the utilities' opportunities to earn the authorized ROE.

Q. Apart from the fully forecasted test years and the ability to earn the allowed return, what other benefits does the RRA profile mention regarding New York State regulation?

A. Apart from the fully forecasted test years and

the ability to earn the allowed return, the profile notes that New York State regulation includes risk-reducing deferral and true-up mechanisms, and full pass-through of commodity costs to ratepayers.

Q. What is your conclusion about New York's State utility regulatory environment?

A. New York State regulation offers a myriad of benefits to the utilities it regulates. Moreover, New York State has a transparent and predictable approach to setting return on equity. These positive aspects of New York State regulation have gained widespread recognition, including from RRA.

Other Risks

Q. In his testimony, did Companies' witness Nowak note that the Companies are exposed to additional risks?

A. Yes, he did. On pages 56 to 58 of his testimony, Companies' witness Nowak argues that the Companies are exposed to three additional risks that were not captured in his ROE analyses for the Companies. Specifically, he argued that the Companies' requirement to maintain certain

customer service quality metrics and gas safety performance metrics makes them riskier than his proxy group. Also, he contends that the Companies are exposed to risk because of the Climate Leadership and Community Protection Act, or CLCPA, relating to decarbonization policy initiatives in New York.

Q. Does Companies' witness Nowak propose to adjust his 9.8 percent proposed ROE to account for the Companies' perceived risks relating to customer service quality and gas safety measures as well as CLCPA?

A. No, he does not. Companies' witness Nowak only discusses these three perceived additional risks but declined to propose a numerical risk premium be added to his ROE results. Instead, he claims that the three additional risks should be noted to support the Companies' requested ROE of 9.8 percent.

Q. What is your response to Companies' witness Nowak's assertion and concern that service quality performance and gas safety benchmark requirements could expose the Companies to additional risk?

A. While it is understandable that the Companies could potentially face additional risks associated with service quality performance and gas safety benchmarks, it is important to consider the broader context. These performance benchmark requirements are not unique to the Companies or their peers in New York; rather, they represent industry-wide standards generally imposed by utility regulatory bodies to ensure reliable, safe, and satisfactory services for customers. The penalties or consequences for failing to meet these benchmarks are intended to incentivize utilities to maintain high standards and prioritize service quality and safety.

Q. Please summarize Companies' witness Nowak's perspectives on the decarbonization policy efforts of the CLCPA and its impact on the Companies' business risk.

A. Companies' witness Nowak claims that the CLCPA, which aims at reducing the use of natural gas, increases the risks to natural gas distribution companies in general, including that of the Companies in particular.

Q. What is your perspective on the effect of the

CLCPA on the Companies' risk profiles?

A. It is difficult to assess the full impact of the CLCPA on gas utilities and the Companies, however, the Companies are not likely to be exposed to any CLCPA-related risk during the Rate Year.

Q. Please summarize your opinion about Companies' witness Nowak's claims that the Companies are exposed to risks relating to their performance metrics.

A. The performance metrics for service quality and gas safety are not unique to the Companies. These are industry-wide standards imposed by utility regulatory bodies to ensure reliable, safe, and satisfactory services for customers.

Service Company Asset Recovery Charge Rate of Return

Q. What is the Service Company asset recovery charge?

A. The Service Company asset recovery charge is an expense item allocated to the affiliates, as described on pages 27 and 28 of the KEDNY Capital Structure Panel's testimony, and pages 26 and 27 of the KEDLI Capital Structure Panel's testimony. It relates to shared assets owned or leased by

the Service Company, primarily shared office facilities, information technology equipment, and software. The Service Company charges its affiliates an asset recovery charge based on an overall rate of return on the asset, net of deferred taxes, and booked depreciation expense.

Q. For both KEDNY and KEDLI, what rate of return does their respective Capital Structure Panels propose to use for the Service Company asset recovery charge, and how was it derived?

A. The Companies requested an overall after-tax cost of capital of 7.32 percent. This rate is based on the 9.8 percent ROE previously discussed, the Service Company's stand-alone capital structure consisting of a 50 percent common equity ratio and a long-term debt ratio of 50 percent with an embedded debt cost rate of 4.85 percent.

Q. Do you agree with the Companies' proposal to use the 7.32 percent after-tax rate of return?

A. No, not entirely. While I do not object to using the Service Company's actual debt cost rate of 4.85 percent, I have concerns regarding the proposed 50 percent common equity ratio and the requested return on equity.

Q. What have the Service Company's affiliates requested and obtained for their common equity ratio in the past?

A. The Service Company's affiliates, including Niagara Mohawk Power Corporation, requested the same 50 percent level of common equity ratio in past cases. Yet, with the exception of the Companies' 2019 rate case, the revenue requirement capital structure of 48.0 percent aligned with the asset recovery capital structure of 48.0 percent, which were subsequently recommended in the joint proposals and ultimately adopted by the Commission in such cases. Examples include Cases 16-G-0058 and 16-G-0059 involving the Companies' 2016 gas rate cases, Cases 12-E-0201 and 12-G-0202 involving Niagara Mohawk Power Corporation d/b/a National Grid - Electric and Gas Rates, and more recently in Cases 20-E-0380 & 20-G-0381 involving Niagara Mohawk Power Corporation Electric and Gas Rates.

Q. What common equity ratio and overall rate of return do you recommend for the Service Company asset recovery charge?

A. I recommend utilizing a 48.0 percent common

equity ratio, as shown in Exhibit___(KXD-34). This would result in an overall after-tax cost of capital of 6.89 percent, which is predicated on my recommended 9.1 percent ROE discussed earlier, my recommended 48.0 percent common equity ratio, and a long-term debt cost rate of 4.85 percent.

Conclusion

Q. What is your recommended cost of capital for the Companies?

A. As summarized in Exhibit___(KXD-2), my recommended overall after-tax rate of return for KEDNY is 6.76 percent and 6.72 percent for KEDLI based on a capital structure with a 48.0 percent common equity ratio and a 9.1 percent cost of equity. As I stated earlier, Companies' witness Nowak's recommended cost of equity range of 9.8 percent to 10.7 percent is excessive and the Companies' requested 9.8 percent cost of equity should be rejected. For the Service Company asset recovery charge, I recommend an overall after-tax cost of capital of 6.89 percent instead of the 7.32 percent requested by the Companies.

Q. Does this conclude your testimony at this time?

A. Yes. Thank you.